

CHEMISTRY GRADE 10: THE PERIODIC TABLE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 1.0: Inorganic Chemistry
Sub-Strand	Sub-Strand 1.3: The Periodic Table
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Development of the periodic table (Groups and periods) – Ion formation (oxidation number, valency, radicals) – Formulae and chemical equations – Chemical families: alkali metals, alkaline earth metals, halogens, noble gases and transition elements
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - relate the position of an element in the periodic table to its electron arrangement, - illustrate ion formation of elements, - derive the formulae of compounds, - write balanced equations for chemical reactions, - appreciate the role of electron arrangement in the development of the periodic table.
Core Competencies	– Communication and Collaboration: the learner listens keenly and actively using open questions while discussing with peers the relationship between atomic number, mass number and number of electrons in atoms of elements. – Critical thinking and problem solving: the learner creates new ideas while drawing diagrams of ions of selected elements using dots and/or crosses to illustrate ion formation. – Learning to learn: the learner learns independently by writing balanced equations for chemical reactions. – Citizenship: the learner appreciates contributions made by various scientists on the development of atomic structure and the periodic table.
Core Values	– Peace: The learner shows compassion as they display tolerance while predicting the type of ion formed from the electron arrangement of elements with peers.
Science & Engineering Practices	– Developing and using models: learners build and revise a cumulative model linking electron arrangement to position, ion formation and formulae. – Analyzing and interpreting data: learners examine element card data (atomic number, electron arrangement, reactivity) to find patterns in groups and periods. – Planning and carrying out investigations: learners test the reactivity of selected elements, e.g. observing sodium, calcium and magnesium behaviour via demonstrations or simulations. – Using mathematics and computational thinking: learners balance equations and derive formulae using valencies and oxidation

	numbers. – Constructing explanations: learners explain why elements in the same group behave similarly.
Pertinent & Contemporary Issues (PCIs)	– Socio-Economic and Environmental Issues (social cohesion): as the learner discusses with peers how atoms acquire stability. – Life skills (analytical thinking): as the learner predicts the type of ion formed from a given electron arrangement of an atom.
Career Connections	– Chemist or analytical chemist at KEBS or the Kenya Bureau of Standards: uses periodic trends to test product quality. – Water treatment engineer (e.g. Nairobi City Water): uses chlorine and fluorine compounds. – Agronomist or soil scientist (Kericho, Rift Valley): selects potassium and nitrogen fertilizers. – Pharmacist or medical laboratory technologist: selects compounds for treatments. – Materials and battery engineer: uses sodium and lithium compounds.
Focus for Lessons	Learners investigate how electron arrangement determines an element's position in the periodic table and its chemical family. They then use this to predict ion formation, valency and oxidation numbers, derive formulae and write balanced equations. The unit follows a phenomenon-driven storyline with predict, observe, explain, Driving Question Board and model-building phases in each lesson.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? KICD KEY INQUIRY QUESTIONS: 1. The KICD curriculum design excerpt supplied for sub-strand 1.3 does not list a key inquiry question; the sub-strand is guided by the teacher's phenomenon question: How does electron arrangement influence the arrangement of elements in the periodic table? 2. (The key inquiry question shown in the excerpt, 'How do chemical bonds influence the properties of substances?', belongs to Sub-Strand 1.4 Chemical Bonding and is the forward link from this unit.)
Anchoring Phenomenon	How does electron arrangement influence the arrangement of elements in the periodic table? Learners receive a set of element cards (the first 20 elements) showing symbol, atomic number and properties, but not their positions. They are asked to sort them as chemists did before they knew about electrons. They then observe that elements such as sodium, potassium and lithium (used in batteries and found in Kenyan fertilizers like potassium compounds) react vigorously with water, while neon and argon do not react at all, and that chlorine and fluorine compounds are used in water treatment in Kenyan towns such as Nairobi and Mombasa. It is puzzling that elements with very different masses behave alike and fall into the same column, while neighbouring elements behave very differently. What hidden property of atoms creates these patterns and lets chemists predict the ions and compounds of elements they have never worked with?
Storyline Thread	Lesson 1: Students sort element cards of the first 20 elements and brainstorm how scientists such as Mendeleev and Moseley arranged them; the DQB is launched with the phenomenon. Lesson 2: Development of the periodic table: groups, periods and the role of atomic number; students arrange the first 20 elements into groups and periods. Lesson 3: Writing electron arrangements (s and p notation) and relating them to group and period number. Lesson 4: Chemical families: alkali metals, alkaline earth metals, halogens, noble gases and transition elements; reactivity patterns observed through demonstration or simulation. Lesson 5: Stability of atoms: why atoms lose or gain electrons; the noble gas stable arrangement. Lesson 6: Ion formation: predicting cations and anions from electron arrangements and drawing dot-and-cross diagrams; electron arrangement of ions. Lesson 7: Valency and oxidation numbers inferred from electron arrangement; elements with variable oxidation numbers (e.g. iron,

	copper). Lesson 8: Radicals and deriving formulae of compounds using valencies and oxidation numbers, e.g. NaCl, CaCl₂, Al₂O₃, and fertilizer compounds such as KNO₃. Lesson 9: Writing and balancing chemical equations for simple reactions, e.g. sodium with water. Lesson 10: Final model revision and presentation: students use electron arrangement to explain the phenomenon, predict an unfamiliar element's position, ion and formula, and link forward to chemical bonding.
Supporting Phenomena	– Why potassium and nitrogen fertilizers suit particular soils in the Rift Valley and Kericho tea farms, while sodium and calcium compounds behave differently. – Why chlorine and fluorine compounds are used in water treatment plants in Kenyan towns, and why both are in the same family. – Why lithium and sodium are used in batteries and react in similar ways with water. – Why neon and argon in lighting and welding do not form compounds, yet sodium and chlorine combine to make table salt (sodium chloride) used in ugali and sukuma wiki cooking.

ATTRIBUTION AND LICENSING

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LESSON 1 (40 minutes (one lesson of the 30 lessons in Sub-Strand 1.3)): Sorting the Element Cards and Launching the DQB

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the phenomenon: why do elements with very different masses behave alike and sit in the same column, while neighbouring elements behave very differently? Learners sort the first 20 element cards the way early chemists did, discover that sorting by properties and mass reveals patterns but also leaves puzzles, and open the Driving Question Board (DQB).
Knowledge	Elements can be sorted by properties and by relative atomic mass (A_r) into patterns. Mass alone leaves puzzles: Li (A_r 7), Na (23) and K (39) are very different in mass yet all react vigorously with water, while neighbouring elements such as Na and Mg, or F and Ne, behave very differently. Ne and Ar do not react at all. Chlorine is used to disinfect water in Nairobi and Mombasa. Ar (A_r 40) is heavier than K (A_r 39) although Ar has the smaller atomic number (Z 18 against Z 19).
Skills	Sorting and classifying data (NGSS SEP: Analyzing and Interpreting Data); asking questions and defining problems (writing DQB questions); developing and using models (initial Class Model, the 'Electron Arrangement Model' poster); constructing explanations (initial ideas); communication and collaboration (listening keenly and using open questions in groups).
Attitudes	Curiosity about hidden patterns in matter; tolerance and compassion when peers' ideas differ (Peace); appreciation of the contributions of scientists such as Mendeleev and Moseley to the development of the periodic table (Citizenship); willingness to say 'I do not know yet' (Learning to learn).
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? (Teacher's phenomenon question; the KICD excerpt for 1.3 lists no key inquiry question. 'How do chemical bonds influence the properties of substances?' belongs to Sub-Strand 1.4 and is the forward link.)
Purpose in Storyline	This is the launch lesson and builds on nothing earlier except learners' prior knowledge of atoms from earlier grades (protons, neutrons, electrons). It raises the puzzles that later lessons resolve: Lesson 2 uses the Ar/K mass puzzle to introduce the history from Mendeleev to Moseley and the idea of groups and periods; Lesson 3 writes electron arrangements; Lesson 4 returns to the Li, Na, K observations. Lesson 1 does NOT explain why; it only creates the need to know.
Safety Notes	This lesson uses cards, video or teacher description of reactions, and no learner handling of reactive metals. If Mwalimu Achieng demonstrates the water reactions, only a trained teacher may use a pea-sized piece of sodium or lithium, behind a safety screen, with goggles on, learners at least 2 m away, and a small amount of water in a large trough. Potassium must NOT be demonstrated in school; show it by video or description only. Never touch the metals with fingers; store them under oil. Chlorine gas must not be used; chlorine in water treatment is discussed from information only.

B. LESSON OVERVIEW

Wanjiku, Otieno, Amina and Kipchoge's Grade 10 class in Kisumu receives envelopes of 20 Element Cards (symbol, atomic number Z, A_r and properties, but NOT position). Learners first predict how chemists, before they knew about electrons, would sort the cards, then sort them in groups, record their rules, and notice what works and what does not. Mwalimu Achieng then presents the phenomenon: Li, Na and K, with very different masses, all react vigorously with water, while Ne and Ar do not react at all; chlorine and fluorine compounds treat water in Nairobi and Mombasa; Ar (A_r 40) is heavier than K (A_r 39) but comes before it. Learners explain what they think is happening, launch the DQB with their questions, and draw an initial Class Model, the 'Electron Arrangement Model' poster, that is revised in every lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mendeleev's Periodic Table (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four. Mwalimu Achieng holds up a bag of maize, beans and a few mixed seeds (or draws them) and asks how they would sort them. Learners then see the envelope of 20 Element Cards (H to Ca) showing symbol, Z, A _r and a property, but no position. Individually, for 2 minutes, learners write in their notebooks: 'Before anyone knew about electrons, how might chemists have sorted these 20 elements? What would you use to put them in order and groups? What do you think will be difficult?' They then share with a partner. Each learner also writes a one-sentence prediction: 'I think elements like lithium, sodium and potassium will behave ____ when put in water because ____.'	Say: 'Imagine you are a chemist in 1860. You know about mass and about what elements look like and do, but nothing about electrons. How would you organise 20 elements?' Hold up the maize and beans example and ask: 'How did you sort these? What feature did you choose?' Give WAIT TIME of 10-15 seconds before taking answers. Ask learners to write silently for 2 minutes. Circulate and note 3 different predictions. Cold-call 3 learners (for example Amina, Otieno, Kipchoge) to share a sorting idea and 2 learners to share their water prediction. Say: 'I am not looking for the right answer; I want your honest thinking.' Do not correct predictions. Record 4-5 predictions on the board under 'Our first ideas'.	Eliciting prior knowledge through individual prediction followed by think-pair-share. Writing before talking commits every learner to an idea so they can later see how their thinking changes, and linking sorting seeds to sorting elements connects familiar Kenyan food items to the unfamiliar task, building a bridge to the phenomenon.	Look for: (1) at least 80% of learners have written a sorting criterion (for example mass, colour, metal or non-metal, state) and a water prediction; (2) learners give reasons rather than just guesses; (3) note misconceptions such as 'heavier elements are more reactive' or 'all metals react the same'. Tally the sorting criteria suggested (mass, state, metallic or not) on the board to use in the next phase.
Observe Phase  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mendeleev's Periodic Table Practice Source: CK-12  Search ARES for similar readings	Part A (15 minutes): Groups spread out the 20 Element Cards and sort them in two rounds. Round 1: sort by properties (metal or non-metal, state, how it behaves). Round 2: arrange all 20 in a single line by increasing A _r , then try to make rows and columns of elements that behave alike. They record their rules on a group chart. Part B (10 minutes): Mwalimu Achieng presents evidence (video or safe teacher demonstration of lithium and sodium; description or video only for potassium): Li fizzes steadily and floats; Na melts into a ball, darts about and fizzes fast; K melts, burns with a lilac flame and reacts very vigorously. Learners	Distribute the envelopes and say: 'Your job is to find patterns. There is no single right way yet.' Give 5 minutes for sorting by properties, then say: 'Now put them in order of mass, from H (1) to Ca (40). Look for elements that behave alike. What do you notice?' Circulate and ask each group: 'Which cards did you struggle to place? Why?' Use WAIT TIME of 10-15 seconds after each question. For Part B, show the Li, Na and K reactions and ask: 'What is the same? What is different?' Pause the video after each metal and ask learners to turn and talk for 30 seconds. Then project the data table with Li 7, Na 23, K 39 and Ne 20, Ar 40 side by side and ask: 'Li, Na and K are	Data sorting and pattern-seeking (Analyzing and Interpreting Data) with a Notice and Wonder routine. Learners physically manipulate cards to generate patterns, discover that the mass line works partly but breaks down, and use the 'notice' and 'wonder' structure to separate observation from explanation, which keeps the focus on evidence and creates real cognitive conflict about the hidden property.	Observe whether groups: (1) produce at least one valid pattern (for example metals on the left, gases elsewhere, repeating reactivity); (2) correctly identify that Li, Na, K behave alike despite different masses; (3) notice that Ne and Ar are unreactive; (4) notice a placement difficulty for Ar and K. Look for use of data (Z, A _r) in group talk. A group that only says 'they are different' without citing numbers or behaviours needs a prompt: 'Which numbers show that?' Collect the group charts as evidence.

	compare with data cards showing that Ne and Ar do not react at all. They read short information on chlorine used to disinfect Lake Victoria water in Nairobi and Mombasa, and on fluoride compounds in some water supplies. In a table they record for each element: Z, A_r, behaviour with water.	very different in mass. What do you notice about how they behave?' Cold-call 3 groups. Then ask: 'Ne and Ar do not react at all. What do you notice about their masses and where they might sit?' Cold-call 2 learners. Point out that Ar (A_r 40) is heavier than K (A_r 39) and Ca (A_r 40) equals Ar in mass: ask 'If we line up by mass alone, where should Ar and K go?' Do not resolve the puzzle; say: 'Hold that thought; it will matter.' Safety: remind learners not to touch reactive metals; only the trained teacher handles them behind the screen.		
Explain Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mendeleev's Periodic Table (Flexbook) Source: CK-12  Search ARES for similar readings	Groups discuss and write on their charts: 'What patterns did we find? What puzzles remain? What could be the hidden property of atoms that makes Li, Na and K alike but Na and Mg, or F and Ne, different?' Using their prior knowledge of atoms from earlier grades (protons, neutrons, electrons), they suggest initial explanations in sentence starters: 'I think ____ because ____.' Learners then take part in a short class discussion in which Wanjiku, Otieno, Amina and Kipchoge's group-style ideas are compared. Mwalimu Achieng briefly mentions that scientists such as Mendeleev and Moseley grappled with the same puzzle and asks learners to brainstorm in groups how they might have arranged the elements, without giving the answer. Each learner then writes a 2-3 sentence individual explanation in their notebook: 'What do you think makes elements behave alike?'	Say: 'We have two kinds of evidence: patterns that work and puzzles that do not fit. Let us talk about both.' Ask: 'What does sorting by mass explain? What does it NOT explain?' WAIT 10-15 seconds. Cold-call 4 learners from different groups, asking each to build on the previous answer ('Do you agree or disagree with Kipchoge? Why?'). Push thinking with: 'What might atoms have inside them that is the same in Li, Na and K, even though their masses differ?' Accept 'electrons' or 'protons' as hypotheses but do not confirm or reject: say 'Good idea. Hold it as a hypothesis; we will test it.' Mention in one minute that Mendeleev (Russian chemist, 1869) arranged elements by mass and left gaps, and that Moseley (English physicist, 1913) showed that atomic number is the true basis of order, but do not teach more here, since Lesson 2 develops this. Address misconceptions respectfully (for example 'heavier means more reactive') by referring back to the	Claim-evidence-reasoning discussion with a talk move of 'build on, agree or disagree'. Learners must link claims to the specific data (Li 7, Na 23, K 39 all reactive; Ne, Ar unreactive; Ar 40 versus K 39) so that explanations are anchored in evidence, and they frame a testable hypothesis about a hidden atomic property, building an explanatory need for the lessons that follow.	Check individual explanations for: (1) a claim that links to evidence (for example 'Li, Na, K react alike so something inside atoms must be the same'); (2) recognition that mass alone is not enough; (3) an initial hypothesis about electrons, protons or arrangement. Use a 3-level look-for: Level 1 describes only; Level 2 identifies a puzzle; Level 3 proposes a hidden property with a reason. Record the number of learners at each level on a class list to compare with the end of the unit.

		K, Na and Li data. Insist on peaceful, respectful listening: 'We disagree with ideas, not people.'		
Driving Question Board (DQB) Creation  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mendeleev's Periodic Table Practice Source: CK-12  Search ARES for similar readings	Mwalimu Achieng reveals the Driving Question on a large wall chart: 'How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form?' Learners each write 2 questions on sticky notes, using the puzzles and observations from today (for example 'Why do Li, Na and K react alike when their masses are so different?', 'Why do neon and argon not react?', 'Why is chlorine used to treat water in Nairobi and Mombasa?', 'Why does argon come before potassium although it is heavier?', 'How can chemists predict what a new element will do?'). Groups combine and sort their notes into categories (for example 'patterns and groups', 'why some elements do not react', 'ions and formulae'), then post them on the DQB wall chart. The class chooses 2 or 3 questions to investigate first.	Say: 'Good scientists ask questions. Write what you are still wondering after today. Questions that start with how or why are especially powerful.' Model one question and one improvement ('Why is sodium reactive?' becomes 'Why do sodium and potassium react alike even though they have different masses?'). Give 3 minutes of silent writing with WAIT TIME of 10-15 seconds before groups talk. Walk around and ask: 'Which of your questions connects to a puzzle we saw?' Cold-call 4 groups to read and place a question, asking them to explain where they put it and why. Cluster similar questions together on the board and name the clusters with learners. Say: 'We will return to this board in every lesson, move questions we can answer, and add new ones.' Keep the DQB visible and photograph or copy it for the offline record. Point out the link to the KICD sub-strand: 'This is the key to the whole unit, from how elements are arranged to the ions and formulae they form.'	Driving Question Board (question-asking and clustering). Learners generate and categorise their own questions from evidence, which gives them ownership of the learning path, makes their curiosity visible, and gives a record against which growing understanding is tracked in later lessons.	Count and review the sticky notes: look for at least 2 per learner; questions that refer to specific evidence (Li, Na, K; Ne, Ar; chlorine in Nairobi and Mombasa; Ar and K masses); and questions that ask how or why rather than yes or no. Note which learners ask about electrons or atomic structure. Use a quick check: ask 3 learners 'Which question on the board matters most to you and why?' Identify learners who wrote only factual or off-topic questions for gentle follow-up.
Model Building Phase  VIDEO: Atomic number, mass number, and isotopes <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Groups draw their first version of the Class Model on a large sheet titled 'Electron Arrangement Model'. It shows: (1) the 20 elements as they placed them (a line or grid), (2) arrows for patterns that worked (for example Li, Na, K react alike), (3) question marks for puzzles (Ar and K; Ne and Ar unreactive), and (4) a labelled sketch of an atom with a nucleus and electrons showing 'hidden	Say: 'A model is our best current idea, and it is meant to change. Show what you know and mark clearly what you do not know yet.' Provide sentence starters: 'We think ___ because ___. We are not sure about ___.' Walk around and ask: 'Where does mass show up in your model? Where does something else need to?' Allow WAIT TIME of 10-15 seconds. Cold-call 3 groups to explain one	Initial model building with explicit uncertainty. Drawing a model that shows what is known, what is a question and what is a hidden mechanism forces learners to organise evidence and ideas into one representation (Developing and Using Models), and the model gives later lessons something concrete to revise instead of starting from nothing.	Check each group model for: (1) the elements placed with some pattern shown; (2) at least one evidence-based pattern (Li, Na, K alike); (3) at least one puzzle marked (Ar and K, or Ne and Ar unreactive); (4) a hypothesis about a hidden property inside the atom. Use the exit tickets to sort learners into 'sure of a pattern', 'sure of a puzzle' and 'no clear idea', and plan targeted questions for Lesson

Search ARES for similar videos EXERCISE: Atomic Number Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	property?' as their initial idea of what decides behaviour. They add a short caption: 'What we think decides where an element belongs and how it behaves.' Groups do a short gallery walk (3 minutes) to compare models, and then the class agrees on a single starter version to be posted next to the DQB.	feature of their model during the gallery walk, and ask 2 other learners to say what they would add. Combine the ideas into one class poster, and say: 'We will revise this poster after each lesson. By Lesson 10 we should be able to use it to predict the position, ion and formula of an element we have never worked with.' Close with a quick exit ticket: 'Write one thing you are now sure of, and one thing you are still wondering.' Remind learners of the forward steps: in Lesson 2 we look at how Mendelev and Moseley solved the ordering problem.		2. Look for respectful, tolerant discussion during the gallery walk.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did every learner offer a sorting rule and a prediction before seeing evidence? (2) Which patterns did groups find on their own, and which puzzles (Li, Na, K alike; Ar and K masses; Ne and Ar unreactive) did they spot without help? (3) Which misconceptions appeared (for example 'heavier means more reactive', 'all metals react alike') and how will you revisit them with evidence in Lessons 2 to 4? (4) Was wait time of 10-15 seconds used consistently, and did cold-calling include quieter learners? (5) Did the DQB capture questions about electrons, groups, ions and formulae, and which clusters are missing? (6) Were safety rules followed during the water reaction presentation? (7) Record the number of learners at each explanation level for comparison at the end of the unit. Keep the DQB, group charts and Class Model poster ready for Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We sorted the first 20 Element Cards (H to Ca) by properties and by mass. We saw that Li, Na and K, with very different masses (7, 23, 39), all react with water, with the reaction getting stronger from Li to K, while Ne and Ar do not react at all. We noted that chlorine is used to treat water in Nairobi and Mombasa, and that Ar (A _r 40) is heavier than K (A _r 39) although it comes first.
What did I learn?	Elements can be sorted by properties and by mass into patterns, but mass alone does not explain why some elements of very different mass behave alike while neighbouring elements behave very differently. Mendelev and Moseley faced this puzzle, and something hidden inside atoms must decide where an element belongs and how it behaves.
How does this explain the phenomenon?	Our first ideas: a hidden property of atoms, possibly linked to electrons, may explain why Li, Na and K are alike and why Ne and Ar are unreactive. We do not yet know what it is. We started the DQB with the driving question and our own questions, and drew the first version of the Class Model, the 'Electron Arrangement Model', to be revised in each lesson.

LESSON 2 (40 minutes): From Mendeleev's Gaps to Moseley's Atomic Number: Building Groups and Periods

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners trace how the periodic table was developed from Mendeleev (1869, ordered by atomic mass, left gaps) to Moseley (1913, atomic number), define groups (columns) and periods (rows), and arrange the first 20 elements by atomic number into groups and periods.
Knowledge	By the end of the lesson the learner should be able to: (1) state that Mendeleev in 1869 ordered elements by atomic mass and left gaps for undiscovered elements; (2) state that Moseley in 1913 showed that atomic number is the true basis of order; (3) define a group as a vertical column and a period as a horizontal row; (4) arrange H to Ca by atomic number into groups 1 to 18 and periods 1 to 4, and explain why Ar (A_r 40) comes before K (A_r 39).
Skills	Developing and using models (arranging the Element Cards into a table); analysing and interpreting data (comparing Z and A_r columns); constructing explanations from evidence (why mass ordering fails for Ar and K); engaging in argument from evidence.
Attitudes	Learners appreciate the contributions of scientists such as Mendeleev and Moseley to the development of the periodic table (Citizenship). Learners show tolerance and listen respectfully when peers disagree about placement of cards (Peace). Learners practise analytical thinking (Life skills).
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table? (Today: what is the true basis of the order, and what do rows and columns mean?)
Purpose in Storyline	In Lesson 1, learners sorted the Element Cards by properties and mass and found that mass alone leaves puzzles: Li, Na and K behave alike, while neighbours behave very differently. Lesson 2 resolves part of the puzzle: elements are ordered by atomic number, and the table has columns (groups) and rows (periods). It does not yet explain WHY elements in a column behave alike. That question is carried to Lesson 3, where learners write electron arrangements and look for the hidden property behind groups and periods.
Safety Notes	No hazardous materials. Paper cards only. Keep walkways clear when groups move to the wall chart. Learners should handle scissors, if used for cutting cards, with care.

B. LESSON OVERVIEW

Mwalimu Achieng reminds the class of the Lesson 1 puzzle. Wanjiku, Otieno, Amina and Kipchoge, in a Grade 10 class in Kisumu, had sorted the Element Cards by mass, yet Li (7), Na (23) and K (39) behave alike, and Ne and Ar do not react at all. Learners predict how the first 20 elements should be ordered. They then observe a short historical data set: Mendeleev's 1869 mass-based order with gaps, and Moseley's 1913 finding. They re-arrange the cards by atomic number into 4 rows (periods) and columns (groups), and notice that Ar (A_r 40) must come before K (A_r 39). Learners then explain the meaning of groups and periods, add questions to the DQB, and add the first piece to the Class Model: order is by atomic number, and columns hold elements with similar behaviour (Li, Na, K together; Ne and Ar together). The reason for this similarity is the open question that carries into Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners work in groups of 4 with	Say: 'Last lesson we sorted the	Prediction with reasoning (Predict-	Check that at least 80% of

 VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mendeleev's Periodic Table (Flexbook) Source: CK-12  Search ARES for similar readings	their Element Cards (H to Ca). Task: 'If you must line up all 20 cards in ONE order so that a chemist anywhere in Nairobi, Mombasa or Kisumu could find any element, which number on the card would you use: atomic number Z or relative atomic mass A_r? Write your prediction and one reason.' Each learner then writes a prediction: 'Where will K (Z19, A_r 39) and Ar (Z18, A_r 40) go if I order by mass? If I order by atomic number?' They also predict how many rows and columns the table will need and which cards will sit in the same column as Li and Na, linking back to the phenomenon of elements that behave alike.	cards and found a puzzle: Li, Na and K are very different in mass but behave alike. Today we ask: what is the best number to order elements by?' Distribute cards. Give 3 minutes of silent individual writing before group talk. WAIT TIME: after asking 'Which number would you use, Z or A_r, and why?', wait 10-15 seconds before taking answers. Cold-call 4 learners (e.g. Wanjiku, Otieno, Amina and Kipchoge's group representatives) to share predictions. Record the tally of Z versus A_r on the board without judging. Say: 'Hold on to your prediction. The evidence will tell us who is right.'	Observe-Explain): committing to a written prediction before seeing the evidence activates prior knowledge from Lesson 1 and creates a need to know, so learners compare their ideas with the evidence in the Observe Phase.	learners wrote a prediction with a reason. Look for use of ideas from Lesson 1 (mass sorting leaves puzzles). Note how many chose Z and how many chose A_r, and whether any predicted a problem for Ar and K. Misconception to listen for: 'heavier atoms must always come later.'
Observe Phase  VIDEO: Periodic Table of Elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mendeleev's Periodic Table Practice Source: CK-12  Search ARES for similar readings	Part A (10 minutes): Learners study a Historical Data Sheet. Mendeleev (Russian chemist, 1869) ordered the elements by increasing atomic mass and placed those with similar properties in the same column, leaving gaps for elements not yet discovered and predicting their properties. Moseley (English physicist, 1913) showed that the true basis of the order is atomic number (number of protons). Part B (10 minutes): Using a data table of Z, A_r, group and period for H to Ca, learners arrange their Element Cards on a grid: H and He in period 1; Li to Ne in period 2; Na to Ar in period 3; K and Ca in period 4. Groups are placed in columns: group 1 (H, Li, Na, K), group 2 (Be, Mg, Ca), groups 13 to 18 (B to Ne, Al to Ar), with Li, Na, K aligned in a column and Ne and Ar aligned in group 18 with He. They record observations: the order when sorted by A_r versus Z; the pair Ar (Z18, A_r 40) and K (Z19, A_r 39);	Circulate with the prompts: 'Show me where Li, Na and K landed. What do you notice about their column?' and 'Run your finger along period 3. What changes from Na to Ar?' Point to the Ar and K cards: 'If we order strictly by mass, which comes first? What happens to the pattern?' WAIT TIME: after the Ar/K question, wait 10-15 seconds. Cold-call 3 groups to report where they placed Ar and K and what they saw. Cold-call 2 more learners to name one gap Mendeleev might have left. Stress that a column is a GROUP and a row is a PERIOD. Ask learners to count the rows and columns and note that H and He are in period 1 only. Tell groups that placing a card by atomic number sometimes puts a heavier atom BEFORE a lighter one. Ask: 'What does that tell you about mass as the basis of order?' Do not yet explain why elements in a column behave alike; say: 'Keep that question. Write it on a sticky note for the	Data-pattern analysis with a physical model: handling the cards on a grid lets learners see patterns in rows and columns directly and notice the anomaly (Ar before K), so the evidence challenges the mass-based prediction.	Observable indicators: cards placed correctly in 4 periods and the correct groups (at least 18 of 20 cards correctly placed); Li, Na and K in one column and Ne and Ar in group 18; learners point out that Ar (40) comes before K (39) and give atomic number as the reason. Check the Observation Sheet for accurate recording of A_r and Z. Look for the misconception that 'period' means column.

	and Ca (A_r 40), which has the same A_r as Ar but sits in a different group.	DQB.'		
Explain Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mendeleev's Periodic Table (Flexbook) Source: CK-12  Search ARES for similar readings	Learners write a claim-evidence-reasoning (CER) paragraph in their notebooks: Claim: what is the correct basis for ordering the elements? Evidence: at least two pieces of data (e.g. Ar Z18 A_r 40 comes before K Z19 A_r 39; Li 7, Na 23 and K 39 line up in one column even though their masses differ greatly; Ca A_r 40 equals Ar A_r 40 but is in a different group). Reasoning: why atomic number gives a consistent order while mass does not, and how Mendeleev's gaps and Moseley's work fit together. They then use their table to answer: (a) What is a group? What is a period? (b) Which period and group is Mg in? (c) Which elements are in the same group as Cl? (d) Predict where an element with Z = 21 would be placed next. Pairs share explanations with another pair.	Model one sentence frame: 'I claim that elements are ordered by ____ because the data show ____.' Say: 'Use numbers from your table, not just words.' WAIT TIME: after posing 'Why did Mendeleev's mass ordering mostly work and yet fail for Ar and K?', wait 10-15 seconds, then allow pair talk for 2 minutes. Cold-call 4 pairs; ask a fifth learner to restate a peer's idea ('Who can say that in your own words?'). Clarify with the scientific summary: Mendeleev (1869) ordered by mass and left gaps; Moseley (1913) showed atomic number is the true basis; the modern table uses atomic number. Define: group = vertical column (1 to 18), period = horizontal row. Connect to the phenomenon: 'So Li, Na and K, which are very different in mass, share a column. But we have not yet explained WHY elements in a column behave alike. That is our job in Lesson 3.' Do not teach electron arrangement in detail now.	Claim-Evidence-Reasoning (CER) with peer explanation: structuring an argument from the table data forces learners to connect evidence to the concept of atomic number, and peer restating exposes gaps in reasoning.	Collect 6 notebooks. Success criteria: the claim names atomic number; at least two data points cited correctly (Ar/K; Li, Na, K same group or Ca/Ar same A_r); correct definitions of group and period; correct answers to (b) period 3, group 2, and (c) F (and Br beyond the first 20 elements not required). Look for misconceptions such as 'groups are rows' or 'Moseley replaced Mendeleev's idea entirely' (his gaps and column patterns were still useful).
Driving Question Board (DQB) Creation  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mendeleev's Periodic Table	Learners return to the DQB started in Lesson 1. In groups, they write on sticky notes (1) one thing they now understand (e.g. 'Order is by atomic number, not mass'), and (2) one new question (e.g. 'Why do Li, Na and K sit in one column and react alike?', 'Why does period 2 have 8 elements?', 'What makes Ne and Ar in the same group do nothing?', 'Why is H in group 1 if it is not like the metals?', 'What did Mendeleev predict in his gaps?'). Each group places two notes on the DQB under 'Answered so far'	Say: 'Our wall should show what we know and what we still need to find out.' Provide each group 2 sticky notes. Cold-call 3 groups to read out and justify a question and say where it belongs. WAIT TIME: after asking 'Which of these questions might the number of electrons help us answer?', wait 10-15 seconds, then take 2 responses. Circle or star questions that link to electron arrangement, the phenomenon, and the Li/Na/K and Ne/Ar patterns. Tell learners: 'Next lesson we write electron	Driving Question Board (question tracking): publicly sorting questions into answered and open ones makes learning progress visible and keeps the phenomenon at the centre of the unit.	Count the number of notes that show a new, productive question connecting columns or rows to the behaviour of elements. Look for learners moving from 'what' questions to 'why' questions. At least one question per group should link to the hidden property that explains the patterns. Note groups that only repeat Lesson 1 questions.

Practice Source: CK-12 Search ARES for similar readings	or 'Still wondering', and they sort the notes into categories such as History, Groups and Periods, and Why Elements Behave Alike.	arrangements to test these ideas.' Mark questions answered today (e.g. Why is Ar before K?) with a tick and the lesson number.		
Model Building Phase VIDEO: Periodic Table of Elements Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Isotopes and Atomic Mass <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	At the Class Model poster, 'Electron Arrangement Model', learners add the first revision for Lesson 2. In groups they draw a 4-row grid with the first 20 elements arranged by atomic number, label rows 'Periods 1 to 4' and columns 'Groups 1 to 18', and mark the column containing Li, Na and K and the column containing He, Ne and Ar. They add a note in a different colour: 'Order is by atomic number (Moseley, 1913), not mass (Mendeleev, 1869). Ar (40) comes before K (39).' They also add a labelled question mark box: 'What hidden property makes elements in one column behave alike and elements in one row change gradually?' Each learner copies the model into their notebook and writes one sentence predicting what that hidden property might be.	Say: 'A model is our best explanation so far. It will change as evidence grows.' Ask a volunteer group (e.g. Kipchoge's group) to draw while others direct: 'Where does K go? Where does Ar go? How do you know?' WAIT TIME: after asking 'What is still missing from our model that would explain why Li, Na and K react alike?', wait 10-15 seconds before cold-calling 3 learners. Accept ideas such as 'something inside the atom' and write them as hypotheses without confirming. Link to the next lesson: 'In Lesson 3 we will write electron arrangements for H to Ca and look for a pattern that matches groups and periods.' Ask learners to complete the exit ticket: 'Why is argon placed before potassium in the periodic table even though argon is heavier?'	Cumulative model building: adding evidence-based elements to a shared Class Model, while marking what is unexplained, lets learners see how new evidence refines their explanation of the phenomenon.	Check the model for a correctly labelled grid (periods as rows, groups as columns), Ar before K, and a statement of atomic number as the basis. Exit ticket success criterion: answer states that elements are ordered by atomic number (protons), not mass, with Ar Z18 before K Z19. Record learners whose hypothesis about the hidden property is plausible (for example, something that changes by one from element to element) for follow-up in Lesson 3.

D. TEACHER REFLECTION

Which learners still believe that heavier atoms must come later, and which used the Ar/K data to revise that idea? Did every group correctly identify groups as columns and periods as rows? Which DQB questions already point toward electron arrangement, and how will I use them to launch Lesson 3? Did I keep to the lesson boundaries by leaving the explanation of why elements in a column behave alike for later lessons?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When the first 20 Element Cards were arranged by atomic number into 4 periods (rows) and groups (columns), Li, Na and K fell in one column and He, Ne and Ar in another. Ar (Z18, A_r 40) had to be placed before K (Z19, A_r 39), and Ca (A_r 40) had the same A_r as Ar but sat in a different group.
What did I learn?	Mendeleev (1869) ordered elements by atomic mass and left gaps for undiscovered elements. Moseley (1913) showed that atomic number is the true basis of order, and the modern table uses atomic number. A group is a vertical column and a period is a

	horizontal row.
How does this explain the phenomenon?	Mass ordering fails for Ar and K, but atomic number gives a consistent order. The table places elements that behave alike, such as Li, Na and K, in the same group even though their masses differ. The hidden property that causes these patterns is still to be explained and will be explored in Lesson 3.

LESSON 3 (40 minutes (one lesson; extend to 80 minutes if double period)): Writing Electron Arrangements: Reading Group and Period from s and p Notation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners write the electron arrangements of the first 20 elements in energy-level form (e.g. 2,8,1) and in s and p notation (e.g. 1s ² 2s ² 2p ⁶ 3s ¹), and use them to read the group number and period number of each element. This gives the 'hidden property' behind the Lesson 2 table of groups and periods that the driving question asks about.
Knowledge	By the end of the lesson the learner should be able to: (a) relate the position of an element in the periodic table to its electron arrangement. Learners know that Z = number of electrons in a neutral atom; that energy levels fill in the order 1s, 2s, 2p, 3s, 3p, 4s; that s holds up to 2 electrons and p up to 6; that period number = number of occupied energy levels; that group 1 and 2 = number of outer electrons, and groups 13 to 18 = outer electrons + 10 (He, with 2 outer electrons, is in group 18).
Skills	Writing electron arrangements in two forms; counting outer (valence) electrons; reading group and period from notation; identifying patterns in data (NGSS: Analyzing and Interpreting Data); using a pattern as evidence to construct an explanation (NGSS: Constructing Explanations); Communication and Collaboration: listening keenly and using open questions while discussing with peers.
Attitudes	Appreciate the role of electron arrangement in the development of the periodic table. Show tolerance (Peace) when peers disagree about where an electron goes, and appreciate the contribution of scientists such as Mendeleev and Moseley (Citizenship). Practise analytical thinking (Life skills) when predicting an element's position from its notation.
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table? (Lesson focus: What do the electron arrangements of Li, Na and K have in common, and how can we read group and period from them?)
Purpose in Storyline	Lesson 2 arranged H to Ca by atomic number into groups (columns) and periods (rows) but did not say why an element belongs in a particular column. Lesson 3 supplies the evidence: electron arrangements for H to Ca. Learners discover that group tracks outer electrons and period tracks occupied energy levels, so Li, Na and K (masses 7, 23, 39) share a column because each has 1 outer electron. This sets up Lesson 4 (families and reactivity patterns) and Lesson 5 (stability).
Safety Notes	No hazardous chemicals are used. Use paper, counters (bottle tops or beans) and the Element Cards. Keep the floor clear of dropped counters. Learners work in groups of 4 and must not eat the beans or maize grains used as counters.

B. LESSON OVERVIEW

Mwalimu Achieng reminds the class of the Lesson 2 table: H to Ca arranged by atomic number into groups and periods. Then she asks the puzzle again: why do Li (7), Na (23) and K (39) sit in the same column and all react vigorously with water, while neighbours Na and Mg, or Cl and Ar, behave very differently? Learners first predict what might be the same inside Li, Na and K atoms. In the Observe phase they use counters (beans and maize grains) to build electron arrangements for the first 20 elements, write them in energy-level form (2,8,1) and s and p notation (1s² 2s² 2p⁶ 3s¹), and fill a data table. In the Explain phase they look for patterns linking outer electrons to group and occupied levels to period, and test the rules on Ne, Ar, K and Ca. The DQB is updated with new questions (why do outer electrons matter for reactions? why is a full outer level special?) and the Class Model 'Electron Arrangement Model' is revised to show the first version of the rule: group from outer electrons, period from occupied energy levels.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Writing the Periodic Table (PLIX) Source: CK-12  Search ARES for similar readings	(5 min) Learners recall the Lesson 2 table of groups and periods for H to Ca displayed on the board. Mwalimu Achieng shows the Element Cards for Li (Z3, A_r 7), Na (Z11, A_r 23) and K (Z19, A_r 39), which sit in the same column, and the cards for Na, Mg, Ne and Ar. Individually, learners silently write a prediction in their notebooks: 'The atoms of Li, Na and K must have this in common inside them: _____. Na and Mg differ in this way: _____. They then draw a quick sketch of where they think the electrons of a sodium atom (11 electrons) are placed. They share with a partner (think-pair-share) and a few ideas go on the board. This links directly to the phenomenon: elements with very different masses fall in the same column and react alike.	Say: 'Look at the Lesson 2 table. Li, Na and K are in the same column but their masses are 7, 23 and 39. Something hidden must be the same inside these atoms. Write your prediction silently.' Give WAIT TIME of 10-15 seconds after asking: 'What could be the same inside Li, Na and K atoms?' before taking any answer. Then say 'Turn to your partner and compare for one minute.' Cold-call 3 pairs (for example Wanjiku and Otieno's group, Amina's group, Kipchoge's group, as names in examples only; use your own learners' names) with: 'Tell us your partner's idea, not your own.' Record all ideas on the board WITHOUT judging them as right or wrong. Ask: 'What would you need to see to check this idea?' Circulate and note learners who mention electrons, energy levels or shells.	Think-Pair-Share with prediction recording. Writing a prediction before instruction makes learners' prior ideas visible and gives them a stake in the evidence; hearing a partner's idea first lowers the risk of speaking and builds the habit of listening keenly. Learners revisit these predictions in the Explain phase to confirm, revise or reject them using evidence.	Observable indicators: (1) at least 80% of learners have a written prediction and a sketch before the pair talk; (2) count how many predictions mention electrons or energy levels (note names for follow-up); (3) note misconceptions, e.g. 'same number of neutrons', 'same mass', 'same colour'. Record these on the teacher checklist to return to in the Explain phase.
Observe Phase  VIDEO: Counting valence electrons for main group elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Valence Electrons (Flexbook) Source: CK-12  Search ARES for similar readings	(15 min) In groups of 4, learners use counters (beans for electrons, with a paper 'energy level' diagram of circles for levels 1 to 4) to build the electron arrangement of atoms of H to Ca. Rules on the worksheet: level 1 holds up to 2 electrons (1s); level 2 holds up to 8 (2s then 2p); level 3 filled to 8 in this lesson (3s then 3p); then 4s takes the next electrons for K and Ca. Each learner is given 5 or 6 elements, so the group covers all 20. For each element they record: (i) Z, (ii) energy-level form e.g. Na 2,8,1; (iii) s and p notation, e.g. Na 1s ² 2s ² 2p ⁶ 3s ¹ ; (iv) number of outer electrons; (v) number of occupied energy levels. They copy	Model one example on the board with the whole class: sodium, Z11. Say: 'Sodium has 11 electrons. Watch how I fill 1s with 2, then 2s with 2, then 2p with 6. That is 10. One electron remains, so it goes into 3s.' Write 2,8,1 and 1s ² 2s ² 2p ⁶ 3s ¹ side by side and explain that the numbers in front of s and p are the energy levels and the small superscripts count electrons. Then release the groups. Circulate and ask: 'Which electrons are in the outermost energy level?' and 'How many energy levels have electrons in them?' Prompt struggling groups: 'How many electrons does the 2p sub-level hold when full?' Use WAIT TIME of 10-15 seconds	Modelling with physical representations (counters on level diagrams), moving from concrete to symbolic. Learners physically place electrons level by level, then translate to the 2,8,1 and s and p forms side by side, which shows that both notations describe the same arrangement. Building every element from the same filling rules turns individual results into a data set that can be searched for patterns (NGSS: Developing and Using Models; Analyzing and Interpreting Data).	Observable indicators: learners correctly build Na as 2,8,1 and 1s ² 2s ² 2p ⁶ 3s ¹ without help; the total of the superscripts equals Z for each element (self-check); the counters on the paper diagram match the written notation; groups agree in the neighbour-group check. Spot-check Cl (2,8,7; 1s ² 2s ² 2p ⁶ 3s ² 3p ⁵) and K (2,8,8,1; ...3p ⁶ 4s ¹). Use a 3-finger self-rating (3 = confident, 2 = need hints, 1 = lost) at the end of the phase and note the learners showing 1.

	the Lesson 2 group and period from the table next to their results. They complete a class data table: H 1s¹; He 1s²; Li 1s² 2s¹; Be 1s² 2s²; B 1s² 2s² 2p¹; C 2p²; N 2p³; O 2p⁴; F 2p⁵; Ne 2p⁶; Na 3s¹; Mg 3s²; Al 3s² 3p¹; Si 3p²; P 3p³; S 3p⁴; Cl 3p⁵; Ar 3p⁶; K 4s¹; Ca 4s². Groups compare results with a neighbouring group to check agreement.	after each question before giving a hint. Cold-call 2 learners per group to read out their group's results for Li, Na and K. Say: 'Place your finger on the outer electrons of Li, Na and K. What do you notice?' Stop at 12 minutes and ask each group to check the Ne, Ar, K and Ca rows carefully, because they are the ones used later in the Explain phase. Note the common errors: putting K's last electron in 3d or 3p, or writing 2p before 2s.		
Explain Phase  VIDEO: Valence Electrons and the Periodic Table - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Valence Electrons Practice Source: CK-12  Search ARES for similar readings	(10 min) Using the completed data table, groups look for patterns. Task 1: highlight the outer electron count for Li, Na and K, and for Be, Mg and Ca, and compare it with the group number from the Lesson 2 table. They find group 1 elements have 1 outer electron and group 2 elements have 2. Task 2: look at B to Ne and Al to Ar. Outer electrons are 3 to 8 and groups are 13 to 18, so group = outer electrons + 10 (He has 2 outer electrons but is in group 18, with a full 1s²). Task 3: count occupied energy levels for H and He (1), Li to Ne (2), Na to Ar (3), K and Ca (4), and compare with the period number. Task 4: test the rules with a 'mystery atom': 2,8,6 (1s² 2s² 2p⁶ 3s² 3p⁴), which gives group 16, period 3 and identifies it as sulfur (S). They then write an explanation: 'Li, Na and K are in the same group because ____; Na is in period 3 because ____.' They revisit their Predict-phase ideas and say which were supported by evidence. Finally they discuss: 'Ne (2,8) and Ar (2,8,8) both have a full outer level and do not react at all. What might that mean?' (the stability idea is developed in Lesson 5, so they	Say: 'Look only at the data table. What pattern do you see between outer electrons and group number?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners for Task 1 and 3 for Task 3, using the question: 'What is your evidence from the table?' Press for evidence: 'Which rows of the table support what you just said?' When learners say 'group number equals outer electrons', ask: 'Does that work for chlorine, which has 7 outer electrons and is in group 17?' to prompt them to refine the rule for groups 13 to 18. Point out the K and Ar mass puzzle from Lesson 2: 'Atomic number still decides the order, and now we see why: each step in Z adds one electron, so the arrangement changes step by step.' Use the mystery atom as a quick whiteboard check: all learners show their answer at once, then cold-call 2 to explain their reasoning. Link back to the phenomenon: 'Li, Na and K have different masses but the same outer electron count. Does this help to explain why they sit in the same column?' Avoid saying why they react the same; say: 'We will test in the next lesson whether	Pattern-seeking with claim-evidence-reasoning (CER). Learners state a claim (e.g. 'group number is linked to outer electrons'), cite specific rows from the data table as evidence, and give reasoning for why the pattern holds across several elements. Applying the rule to a mystery atom tests whether the pattern is general. This builds the cause-and-effect link between electron arrangement and position in the periodic table (NGSS: Constructing Explanations; Patterns).	Observable indicators: learners cite specific data (for example 'Na is 2,8,1 with 1 outer electron and it is in group 1') rather than asserting the rule; they correctly identify the mystery atom as sulfur in group 16, period 3; they explain why He is placed in group 18 despite having 2 outer electrons; they correctly state that K is in period 4 because it has electrons in 4 energy levels. Collect a written exit sentence from each learner: 'Li, Na and K are in the same group because ____.' Look for 'same number of outer electrons' as the key idea. Misconception to watch for: confusing the number of electrons in total with the number in the outer level.

	only record the pattern for now).	outer electrons also explain how they react.'		
Driving Question Board (DQB) Creation  VIDEO: Valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Counting valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(5 min) Learners return to the Driving Question Board (the class wall chart started in Lesson 1). Individually they write one new question on a sticky note based on today's evidence, for example 'Why do atoms with 1 outer electron react with water?', 'Why are atoms with a full outer level, like Ne and Ar, unreactive?', 'Why does Mg (2 outer electrons) behave differently from Na (1 outer electron)?', 'Do all elements follow the group = outer electrons rule? What about iron and copper?' Then they place their notes under suitable headings (Pattern in the table, Reactivity, Ions and compounds). They also tick questions already answered, such as 'What is the same inside Li, Na and K atoms?'	Say: 'What new question has today's data made you ask? Write it so that someone from another group can understand it.' Give WAIT TIME of 10-15 seconds before learners start writing. Select 3 notes at random and cold-call the authors to read them aloud; ask the class: 'Which lesson do you think might help us answer this?' Group similar questions together, and point out that questions about reactivity and full outer levels will be addressed in Lessons 4 and 5, and questions about ions and formulae later in the sequence. Mark the question 'What is the same inside Li, Na and K atoms?' as 'partly answered: same outer electron count'. Do not answer the new questions yet.	Question generation and sorting on a Driving Question Board. Writing questions from evidence keeps learners' own curiosity driving the unit and shows how today's pattern leads to the next investigation. Sorting questions into categories helps learners see the storyline of the whole unit.	Observable indicators: each learner posts at least one question; at least half of the questions refer to outer electrons, full outer levels, or reactions (showing that learners link electron arrangement to behaviour); questions use today's vocabulary (energy level, outer electrons, group, period). Note learners whose questions are only about copying notation, and follow up with them individually.
Model Building Phase  VIDEO: Groups of the periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Electron configurations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(5 min) Learners gather around the Class Model poster 'Electron Arrangement Model'. As a class they add a new section: 'Inside the atom: electrons are arranged in energy levels (1s; 2s 2p; 3s 3p; 4s).' They add two rules to the poster: 'Group number (groups 1, 2) = number of outer electrons; groups 13 to 18 = outer electrons + 10' and 'Period number = number of occupied energy levels'. They draw Li, Na and K as three level diagrams (2,1; 2,8,1; 2,8,8,1 for K) with the outer electron coloured in, and add a note: 'Same outer electrons, same group, different mass'. Each group also adds a labelled sketch of an atom of their choice (for example Cl, 2,8,7, in group 17, period 3 for the water treatment in Nairobi and	Say: 'Our model must explain the pattern we found. What belongs on the poster, and what evidence supports it?' Give WAIT TIME of 10-15 seconds, then cold-call 4 learners, one for each item to be added, asking 'What is your evidence?' before adding anything to the poster. Ask: 'Which part of the phenomenon can our model already explain, and which part can it not yet explain?' Guide learners to see that it explains the column pattern (Li, Na, K) and the row pattern (period), but not yet the reactivity of Li, Na and K or the lack of reaction of Ne and Ar. Write these gaps in the 'still need to explain' box. Briefly connect to the history: Moseley used atomic number to order elements; today we see that atomic number sets	Cumulative model revision (NGSS: Developing and Using Models). Learners add new evidence-based components to a shared model and explicitly list what the model cannot yet explain, which sets the purpose for Lessons 4 and 5. A visible, revisable model helps learners see how each lesson builds on the previous one toward explaining the driving question.	Observable indicators: the exit slip shows Cl as 1s2 2s2 2p6 3s2 3p5, group 17, period 3 (or 2,8,7); learners can point to the poster and explain why Li, Na and K are in the same group; the poster includes the group and period rules and level diagrams; learners identify at least one thing the model cannot yet explain. Collect exit slips and sort them into three piles (correct, minor error, needs reteach) for planning the start of Lesson 4.

	Mombasa). They add a 'What we still need to explain' box: why outer electrons affect how an element reacts, and why the full outer levels of Ne and Ar matter.	the number of electrons and hence the arrangement. Close with the exit slip: 'Write the s and p notation for chlorine and state its group and period.'		
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners generate the group and period rules from the data table, or did they only copy the rule from you? Which questions prompted real reasoning? (2) Which notation errors were most common (for example putting K's last electron in 3p or 3d, writing 2p before 2s, or forgetting that total superscripts must equal Z)? Plan a quick correction for the start of Lesson 4. (3) Did learners confuse 'group = outer electrons' with groups 13 to 18 (outer electrons + 10), and did He cause confusion? (4) Which learners hesitated to speak in the Think-Pair-Share, and how can you give them a role next time? (5) Were the Kenyan contexts (Kisumu classroom, chlorine in Nairobi and Mombasa, potassium in Kericho farms) used to keep the phenomenon in view? (6) What do the exit slips and DQB questions tell you about readiness for Lesson 4 on families and reactivity? Note any learners who need extra support with counting electrons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built the electron arrangements of H to Ca using counters and wrote them in energy-level form (e.g. Na 2,8,1) and s and p notation (e.g. Na 1s ² 2s ² 2p ⁶ 3s ¹). We noticed that Li (2,1), Na (2,8,1) and K (2,8,8,1) each have 1 outer electron, and that Ne (2,8) and Ar (2,8,8) have full outer levels.
What did I learn?	Group number is decided by the number of outer electrons (group 1 and 2 = number of outer electrons; groups 13 to 18 = outer electrons + 10, with He, 1s ² , in group 18). Period number equals the number of occupied energy levels. Elements in the same group have the same outer electron count even though their masses differ.
How does this explain the phenomenon?	Li, Na and K sit in the same column despite masses of 7, 23 and 39 because each atom has 1 outer electron, while neighbours such as Na and Mg differ because their outer electron counts differ (1 and 2). Electron arrangement, the hidden property that follows from atomic number, therefore decides where an element is placed in the periodic table. We still need to explain why outer electrons control how elements react.

LESSON 4 (40 minutes (one lesson; Predict 5 min, Observe 12 min, Explain 10 min, DQB 5 min, Model 8 min)): Chemical Families and Reactivity Patterns: Why Li, Na and K Behave Alike

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners name the five chemical families, locate them by group, and use outer electron count to explain why elements in a family react alike and why reactivity changes down a group, taking us closer to answering how electron arrangement influences the arrangement of elements in the periodic table.
Knowledge	Alkali metals (group 1) have 1 outer electron; alkaline earth metals (group 2) have 2; halogens (group 17) have 7; noble gases (group 18) have a full outer level (8; He has 2); transition elements lie between group 2 and group 13 from period 4 onward and include Fe and Cu. Reactivity of alkali metals with water increases down the group: $\text{Li} < \text{Na} < \text{K}$. Mg and Ca are in group 2, with Ca more reactive than Mg. Ne and Ar do not react. Cl_2 (and fluoride compounds) are used in water treatment in Nairobi and Mombasa.
Skills	Observing and recording a reaction (demonstration or simulation); analysing and interpreting data (NGSS SEP: Analyzing and Interpreting Data); identifying patterns (Crosscutting: Patterns); constructing explanations using evidence (SEP: Constructing Explanations); communicating and collaborating in groups using open questions.
Attitudes	Learners appreciate the role of electron arrangement in the development of the periodic table; show tolerance and compassion when peers' predictions differ (Peace); work with peers respectfully on shared evidence (Citizenship, social cohesion); apply analytical thinking (Life skills) to patterns in data.
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table? (Sub-strand guiding question from the teacher's phenomenon; the KICD question 'How do chemical bonds influence the properties of substances?' belongs to 1.4 Chemical Bonding and is the forward link.)
Purpose in Storyline	Lesson 3 gave learners the rule that group number comes from outer electrons and period from occupied energy levels. Lesson 1 left the puzzle that Li, Na and K are alike despite different masses. Lesson 4 names the families, gathers reaction evidence, and links family behaviour to outer electron count. It sets up Lesson 5 (why atoms lose or gain electrons to reach a full outer level), which explains the noble gases' inertness.
Safety Notes	Preferably use a teacher-led video or simulation of Na and K with water. If a live demonstration is done: only the teacher handles the metals with forceps; use a rice-grain-sized piece of Na (and never K live); use a large trough half-filled with water behind a clear screen; learners stand at least 2 metres back and wear safety goggles; keep metals stored under oil, dry cut pieces on filter paper, and dispose of leftovers as the teacher directs. Li may be shown in a very small piece. Do not let learners touch alkali metals. Chlorine gas is NOT demonstrated; use photographs and a reading about water treatment. Wash hands after handling any materials. Have a bucket of sand and a fire blanket ready, and never use water on a metal fire.

B. LESSON OVERVIEW

Mwalimu Achieng opens by asking learners to revisit the Element Cards and the Lesson 1 puzzle: Li (A_r 7), Na (23) and K (39) are very different in mass yet all react vigorously with water, while Ne and Ar do not react at all. Learners predict the order of reactivity and what feature of the atoms makes them alike. They then gather evidence from a teacher demonstration or simulation of Li, Na and K with water, and from data cards on Mg and Ca, Ne and Ar, Cl and F (water treatment in Nairobi, Mombasa and Lake Victoria supplies), and Fe and Cu (transition elements). Groups of four (Wanjiku, Otieno, Amina and Kipchoge are example names) build a Family Table linking each family to group number and outer electrons from Lesson 3. In the Explain phase they construct a claim-evidence-reasoning statement that family

behaviour follows outer electron count. New questions are added to the Driving Question Board (DQB), and the Class Model (Electron Arrangement Model) is revised to add families and the idea that outer electrons decide behaviour. Why atoms lose or gain electrons is deliberately left to Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Alkaline Earth Metals Practice Source: CK-12  Search ARES for similar readings	Learners look at the Element Cards for Li, Na, K, Ne and Ar (symbol, Z, A_r and properties, but no positions). Individually, in 2 minutes, they write on a sticky note or in their notebooks: (1) the order in which Li, Na and K will react with water from slowest to fastest, and (2) one reason why Li, Na and K, with masses 7, 23 and 39, might behave alike. They use Lesson 3 to write the outer electron count of each (Li 2,1; Na 2,8,1; K 2,8,8,1) and predict what Ne (2,8) and Ar (2,8,8) will do. They pair-share and draw a quick sketch of what they expect to see when a piece of sodium meets water.	Display the five cards on the board. Say: "In Lesson 1 we saw that Li, Na and K behave alike even though their masses are very different. Today we find out what family they belong to and whether we can predict who reacts fastest. Write your prediction first, quietly." Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds after asking: "What do Li, Na and K have in common according to the electron arrangements you wrote in Lesson 3?" Then cold-call 4 learners (e.g. Wanjiku, Otieno, Amina, Kipchoge) for predictions without judging them, saying: "Thank you, I am recording that on the board. We will test it with evidence." Tally the predicted orders on the board. Accept all predictions and do not reveal the answer. Remind the class to respect different views (Peace).	Predict-Observe-Explain (prediction stage) with a Think-Pair-Share: committing to a prediction in writing exposes prior ideas, such as 'heavier metals react faster', so that later evidence can confirm or challenge them. It links to the phenomenon by asking what hidden property makes elements of different masses alike.	Check: (a) at least 80% of learners wrote a predicted order and a reason; (b) learners correctly state outer electron counts for Li, Na and K as 1 from their Lesson 3 arrangements; (c) listen for misconceptions such as 'mass decides reactivity' or 'neon is just a slow reactor'; note names and tally the types of prediction on the board for use in the Explain phase.
Observe Phase  VIDEO: The periodic table - classification of elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Alkaline Earth Metals (Flexbook)	Learners gather evidence in three stations of 4 minutes each (or as a whole-class sequence if resources are limited). Station/Part A: watch the teacher demonstration or video/simulation of Li, Na and K in water. They record in a table: does it float or sink, melt, fizz (rate), flame colour, and relative speed. Expected qualitative observations: Li fizzes steadily and floats; Na melts into a ball, darts and fizzes fast; K melts, burns with a lilac flame and is very vigorous. They	Set safety rules first: "Everyone stays behind the line, goggles on, no touching the metals." If demonstrating, cut each piece at the front and say aloud: "Watch the metal's surface, the speed, and any flame. Record exactly what you see, not what you expected." Run the sequence Li, then Na (and show K by video only). Pause after each: "What did you observe? Describe it without explaining it yet." WAIT TIME of 10-15 seconds before taking answers. Cold-call 3	Data table and pattern-finding (Analyzing and Interpreting Data; Patterns): recording observations in a Family Table, separate from explanations, lets learners see that reactivity increases down group 1 and that behaviour tracks the column, not the mass. Using a demonstration/simulation and data cards provides multiple sources of evidence.	Observable indicators: learners' tables record float/melt/fizz/flame for Li, Na, K accurately and in the order $Li < Na < K$; they label at least four families correctly with group numbers; they write 1, 2, 7 and full for outer electrons in the correct rows; they state that Ne and Ar showed no reaction. Walk around with a checklist; ask any group with a wrong family-to-group match: "Check your Lesson 3 arrangement again. How many outer electrons does it have?"

Source: CK-12  Search ARES for similar readings	test a drop of universal indicator or red litmus in the water after to note it turns alkaline (the products are LiOH, NaOH, KOH solutions, with the equation left for Lesson 9). Part B: Data cards. Card 1: Mg and Ca with water or dilute acid: Ca reacts faster than Mg; both are group 2 with 2 outer electrons (Mg 2,8,2; Ca 2,8,8,2). Card 2: Ne and Ar: no reactions; they have full outer levels (2,8 and 2,8,8). Card 3: Cl₂ and fluoride compounds in water treatment in Nairobi and Mombasa and in water drawn from Lake Victoria in Kisumu: halogens have 7 outer electrons (F 2,7; Cl 2,8,7) and are reactive non-metals. Card 4: Fe and Cu, transition elements from period 4 (Fe rusts in Kisumu's humid air and in mabati roofs; Cu is used in electrical wires), which form more than one ion (Fe²⁺ and Fe³⁺, Cu⁺ and Cu²⁺), a pattern not predicted by outer electron count alone, to be explored in Lesson 7. Learners complete a Family Table with the columns: family, group, example elements, outer electrons, behaviour observed.	learners per metal to give one observation each. For Part B, circulate and ask groups: "Which of these elements react and which do not? What do the reactive ones in the same column share?" and "How many outer electrons does each have?" Prompt for the families by name: "We call group 1 the alkali metals, group 2 the alkaline earth metals, group 17 the halogens, group 18 the noble gases, and the block between group 2 and 13 from period 4 the transition elements." Collect the observations on a class chart, ordering Li < Na < K for reactivity. Point out that Ar (A_r 40) and Ca (A_r 40) have the same mass but opposite behaviour (Ar does not react, Ca does), reinforcing the Lesson 1 puzzle.		
Explain Phase  VIDEO: Oxidation state trends in periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Modern Periodic Table: Periods	In groups of four, learners use the Family Table to write a claim-evidence-reasoning (CER) statement: Claim: elements in the same family behave alike because they have the same number of outer electrons. Evidence: Li (2,1), Na (2,8,1) and K (2,8,8,1) all have 1 outer electron and all react with water, though their masses are 7, 23 and 39; Ne (2,8) and Ar (2,8,8) have full outer levels and do not react. Reasoning: outer electrons decide how an atom behaves, so the group number tells us the family. They then answer: (1) Why	Say: "Use only your evidence to complete the claim. Every reason must point to outer electrons." Circulate and ask: "Which observation supports your claim?" and "Does mass explain it? Look at Ar 40 and Ca 40." Use talk moves: "Who can add to what Amina said?" and "Do you agree or disagree with Otieno, and why?" Give WAIT TIME of 10-15 seconds after each question. Cold-call 5 learners from different groups to share one part of their CER, and ask the class to give a thumbs up or down on whether the evidence	Claim-Evidence-Reasoning with Talk Moves (Constructing Explanations and Engaging in Argument from Evidence): learners must link a general claim to specific data (Li, Na, K; Ne, Ar; Mg, Ca) and to the concept of outer electrons, building a causal chain from the arrangement of electrons to the behaviour of a family. Predicting Rb's reactivity tests whether the pattern transfers to an unseen element, a step toward the Lesson 10 strontium task.	Look for: CER statements that cite at least two pieces of data and mention outer electrons in the reasoning; correct answer that Na and Mg differ because of 1 versus 2 outer electrons; correct prediction that Rb is more reactive than K; no claims that mass causes the similarity. Use a quick exit check: "Complete: Elements in group 1 react alike because ____." Learners who say 'they have similar mass' are grouped for a short re-teach using the Element Cards while others extend with Fr or Cs.

and Groups (Flexbook) Source: CK-12 Search ARES for similar readings	are Na and Mg, neighbours in period 3, so different? (Na has 1 outer electron, Mg has 2.) (2) Why do Na and Ne, with similar masses 23 and 20, behave so differently? (3) Predict: rubidium (Rb) is below K in group 1; will it be more or less reactive than K, and why? (More reactive; same outer electron count, trend continues down.) (4) Chlorine is in group 17: how many outer electrons, and why is it classed with fluorine? Learners state that they do not yet know why the number of outer electrons causes such behaviour, and write this as a question for the DQB (this is left for Lesson 5). They compare their Lesson 4 predictions with the evidence and revise them.	supports it. Address the misconception that heavier means more reactive across all groups: "Within a group, reactivity increases down the group; across a period, neighbours differ because their outer electron counts differ." Do NOT explain why atoms lose or gain electrons or the stability of full outer levels; say: "That is a great question and Lesson 5 will answer it. Put it on the DQB." Note that for transition elements, Fe and Cu have variable ions that outer electron count alone does not predict; this is explored in Lesson 7.		
Driving Question Board (DQB) Creation  VIDEO: Atomic radius trends on periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  PDF: Periodicity of Chemical families Source: ARES Local Curriculum - Digital Literacy Content Search ARES for similar readings	Learners add new questions and partial answers to the class DQB wall chart started in Lesson 1. They place sticky notes under categories such as 'Answered today', 'New questions' and 'Still puzzling'. Example answered items: 'Why do Li, Na and K behave alike?' (same outer electron count, 1). Example new questions: 'Why do atoms with full outer levels not react?', 'Why does the reactivity of group 1 metals increase down the group?', 'Why do halogens react in the opposite way to alkali metals?', 'Why do iron and copper form more than one kind of ion?', 'Why is chlorine used in water treatment in Nairobi and Mombasa?' Each learner writes at least one question, and pairs decide which lesson might answer it.	Say: "Look at the board. What have we answered about our driving question today? What are we still wondering?" Give WAIT TIME of 10-15 seconds, then invite 4 learners to read a sticky note aloud and place it. Ask: "Which question connects to what we saw in the sodium reaction?" Move 'Why do Li, Na and K behave alike?' to a 'Partly answered' column and label it 'same outer electrons, Lesson 4'. Tell the class that the 'why full outer levels are stable' question will be tackled in Lesson 5, and the 'Fe and Cu' question in Lesson 7. Praise good questions with a comment such as: "That question will help us explain the phenomenon."	Driving Question Board as a public record of growing understanding: connecting learner questions to evidence and to future lessons builds ownership and shows that science progresses by turning observations into new questions.	Count questions: every learner contributed at least one note; at least half of the notes refer to why or how (causal) rather than what; check that learners correctly sort notes into 'answered' versus 'new'. Note any learner whose question reveals a misconception, such as 'noble gases are unreactive because they are light', and revisit it in Lesson 5.
Model Building	Learners revise the class poster, 'Electron Arrangement Model'. In	Say: "Take the model from Lesson 3. What must we add today, based	Cumulative Model Revision (Developing and Using Models):	Check that the revised model includes the correct family-group-

Phase  VIDEO: Counting valence electrons for main group elements <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hydrogen and Alkali Metals (Flexbook) Source: CK-12  Search ARES for similar readings	groups, they add to the model: (1) the five family names with group numbers (alkali metals group 1, alkaline earth metals group 2, transition elements between group 2 and 13 from period 4, halogens group 17, noble gases group 18); (2) outer electron counts of 1, 2, 7 and full for the four main families; (3) an arrow showing reactivity of alkali metals increasing down group 1 (Li < Na < K); (4) the statement 'Same outer electrons = same family behaviour, regardless of mass'. They also add the label 'Mass does not decide behaviour: Li 7, Na 23, K 39 alike; Na 23 and Ne 20 differ'. Each group draws a mini-diagram of the electron arrangement of one family member, e.g. 2,8,1 for Na and 2,8,7 for Cl, and notes with a question mark the part not yet explained (why outer electrons matter). Individually, learners complete the three-column summary table in their notebooks: What we observed, What we learned, How we can explain it.	on evidence?" Give WAIT TIME of 10-15 seconds, then cold-call 3 groups to propose one addition each and ask the class: "Is there evidence on our chart to support that?" Insist that every addition cite an observation. Mark unexplained parts of the model with a question mark. Ask: "Where in our model does the Rb prediction fit?" Close with: "We now know that the outer electrons give elements a family identity. Next lesson we ask why outer electrons make an atom react or not react." Collect the summary tables as an exit ticket.	updating the Class Model with only evidence-based additions, and marking gaps, shows how a model is improved over time and connects the family patterns to the underlying cause, electron arrangement.	outer-electron links and a reactivity trend arrow; check at least 80% of the learners' tables state that outer electrons, not mass, explain family similarity; note any learner who cannot link Ne and Ar to a full outer level. Use the answers as an exit ticket to plan the opening of Lesson 5.
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D. TEACHER REFLECTION

After the lesson, reflect: Did learners rely on the evidence (Li, Na, K reactions; Ne and Ar; Mg and Ca) rather than on mass to explain the family patterns? Which learners still said 'heavier reacts faster'? Was the safety plan followed, and was the demonstration or simulation visible to everyone in the class? Did I resist explaining why full outer levels are stable, leaving it for Lesson 5? Which DQB questions should open Lesson 5? How well did Kisumu-based contexts such as Lake Victoria water treatment, mabati roofs and Kericho fertilizers help learners connect with the content? What would I change next time about wait time, cold-calling and group roles?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Li fizzes steadily and floats; Na melts into a ball, darts and fizzes fast; K melts, burns with a lilac flame and is very vigorous. Ca reacts faster than Mg. Ne and Ar do not react. Cl and F compounds are used in water treatment in Nairobi and Mombasa. Fe and Cu form more than one ion.
What did I learn?	Elements fall into families: alkali metals (group 1, 1 outer electron), alkaline earth metals (group 2, 2 outer electrons), halogens

	(group 17, 7 outer electrons), noble gases (group 18, full outer level) and transition elements (between group 2 and 13 from period 4). Reactivity of alkali metals increases down the group: $\text{Li} < \text{Na} < \text{K}$.
How does this explain the phenomenon?	Elements in the same family behave alike because they have the same number of outer electrons, not the same mass (Li 7, Na 23 and K 39 all have 1 outer electron). Neighbours such as Na and Mg or Na and Ne differ because their outer electron counts differ. Why a full outer level makes Ne and Ar unreactive is still to be explained in Lesson 5.

LESSON 5 (40 minutes): Stability of Atoms: Why Atoms Lose or Gain Electrons

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use outer electron counts of He, Ne, Ar, Na, Mg, O and Cl to explain why noble gases are unreactive and why other atoms lose or gain electrons to reach a full outer level.
Knowledge	By the end of the lesson, the learner should be able to: (1) state that noble gases (He 1s ² ; Ne 2,8; Ar 2,8,8) have full outer energy levels (duplet for He, octet for Ne and Ar) that are stable; (2) explain that atoms with few outer electrons (Na 1, Mg 2) tend to lose electrons, and atoms with nearly full outer levels (O 6, Cl 7) tend to gain electrons, to reach the noble gas arrangement; (3) relate stability to the phenomenon of why group 1 metals react vigorously while Ne and Ar do not. Linked KICD outcome: relate the position of an element in the periodic table to its electron arrangement (a); and discuss with peers the stability of atoms (loss or gain of electrons).
Skills	Developing and using models (NGSS SEP 2): sketch energy-level diagrams and use them to predict whether an atom will lose or gain electrons. Constructing explanations (SEP 6): use evidence of outer electron counts and reactivity to explain stability. Engaging in argument from evidence (SEP 7): defend a claim about the easiest route to a full outer level. Communication and Collaboration: listen keenly and use open questions in group discussion.
Attitudes	Appreciate the role of electron arrangement in explaining patterns in the periodic table. Show tolerance and compassion (Peace) by listening to and respectfully challenging peers' predictions. Show social cohesion (PCI) by sharing roles and ideas fairly in groups.
Key Inquiry Question	Why do some atoms, such as Na and Cl, readily lose or gain electrons, while Ne and Ar do not react at all?
Purpose in Storyline	Lessons 3 and 4 showed that outer electron count decides the group and that noble gases (Ne, Ar) are unreactive while Li, Na, K react vigorously. Lesson 5 supplies the missing reason: a full outer level is stable, so atoms with other outer electron counts change to reach it. This sets up Lesson 6, where learners predict the ions formed and draw dot-and-cross diagrams. It advances the driving question by linking electron arrangement to the ions and compounds elements form.
Safety Notes	This lesson is paper and discussion based with no chemicals. Where the teacher demonstrates or recalls the sodium or potassium water reaction, use only a video or the earlier observations; do not handle alkali metals in class. Keep walkways clear during gallery walk movement. Use blunt-tipped counters and avoid throwing them.

B. LESSON OVERVIEW

Mwalimu Achieng reminds the class from Lessons 3 and 4 that Ne and Ar do not react while Na, which has one outer electron, reacts vigorously. Learners at their Kisumu school first predict which atoms will gain, lose or keep electrons. They then gather evidence with an electron-counter card sort: using bean and maize seed counters as electrons, they build He, Ne, Ar, Na, Mg, O and Cl, and count how many electrons must move to make a full outer level. In the Explain phase, learners use the 'Claim-Evidence-Reasoning' strategy to explain why Na loses 1 electron, not gains 7, and why Cl gains 1, not loses 7. They then add new questions to the Driving Question Board (DQB) and revise the Class Model ('Electron Arrangement Model') with the stability idea. This answers part of the phenomenon: why elements in different groups behave very differently, and why neighbours like Na, Mg, Ne and Ar differ.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Counting valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the phenomenon: Li, Na and K react vigorously with water while Ne and Ar do not react at all, and chlorine is used in water treatment in Nairobi and Mombasa. Each learner receives a prediction slip with Ne 2,8; Na 2,8,1; Mg 2,8,2; O 2,6; Cl 2,8,7. On the slip they answer individually: 'Which of these atoms do you think is most stable (least likely to react)? Which will lose electrons, which will gain, and why?' They draw a quick energy-level sketch for one atom and write one sentence of reasoning. Then they pair up with Wanjiku-, Otieno-, Amina- or Kipchoge-style partners to compare and record agreements and disagreements.	Display the Element Cards for Ne, Na, Cl and the DQB. Say: 'Yesterday we saw that neon and argon sit in group 18 and do not react. Today we ask why. Before I teach anything, I want your honest prediction.' Give 3 minutes of silent writing. Ask: 'Look at Na 2,8,1 and Cl 2,8,7. Which one is closer to Ne 2,8 and which is closer to Ar 2,8,8?' WAIT TIME: 10-15 seconds, no hands up. Cold-call 3 learners: 'Amina, what did you predict for sodium? Why?' Record 2-3 different predictions on the board without judging. Say: 'There are no wrong predictions right now; we will test them with evidence.' Circulate and note who thinks atoms 'want' electrons, to revisit in the Explain phase.	Predict-Observe-Explain (prior-ideas elicitation) with Think-Pair-Share: writing a prediction before any instruction commits learners to an idea, so that evidence later either supports or challenges it. It makes their thinking visible and gives the class a shared reference to return to.	Look for: (1) learners who link Ne and Ar's lack of reactivity to full outer energy levels or 8 outer electrons; (2) learners who sketch the correct outer electron count for Na (1), Mg (2), O (6) and Cl (7); (3) learners who give a reason beyond 'because it wants to' (such as 'closer to 8'). Tally on a checklist how many correctly sketch two or more atoms. Note misconceptions (e.g. 'atoms need to be full of electrons in total') for the Explain phase.
Observe Phase  VIDEO: Valence Electrons and the Periodic Table - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Counting valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of four, learners use an Electron Counter Card Sort. Each group receives a mat with energy-level rings and 30 counters (maize seeds as electrons). Step 1: They build He (1s ² , duplet), Ne (2,8) and Ar (2,8,8) and record 'outer electrons: 2, 8, 8' and 'reactivity: none'. Step 2: They build Na (2,8,1), Mg (2,8,2), O (2,6) and Cl (2,8,7). For each, they count (a) how many electrons must be removed to leave a full outer level beneath, and (b) how many must be added to fill the outer level. They complete a table: Atom Outer electrons Electrons to lose Electrons to gain Easier route? Step 3: They compare with the Ne and Ar mats and write what arrangement each atom would end up with (e.g. Na after losing 1 electron: 2,8 like Ne; Cl after gaining 1: 2,8,8 like Ar). They also	Distribute mats and counters. Say: 'You are the chemists in the 1900s who have just learned about electrons. Your job is to collect data, not to guess.' Model one example on the board: build Na 2,8,1 and ask 'What is left if I take away the one outer electron?' WAIT TIME: 10-15 seconds. Circulate with a checklist and ask each group: 'Show me how many moves Mg needs to lose electrons and how many to gain. Which is fewer?' Prompts: 'What do you notice about the arrangements Na, Mg, O and Cl end with? Compare with your Ne and Ar mats.' Cold-call 2 groups to share their table for O and Cl: 'Kipchoge's group, how many electrons does oxygen need to gain? What does it become like?' Remind groups to include Kenyan connections: 'Chlorine in the water of Nairobi	Data table and pattern-finding using physical models (Developing and Using Models, SEP 2): moving counters makes the abstract idea of losing or gaining electrons concrete, and the table helps learners compare fewest-moves routes to spot the pattern that atoms end with the noble gas arrangement.	Observable indicators: (1) group tables correctly show Na: lose 1/gain 7; Mg: lose 2/gain 6; O: lose 6/gain 2; Cl: lose 7/gain 1; (2) learners correctly state that the full outer level reached is 2,8 (Ne-like) or 2,8,8 (Ar-like); (3) learners use counters accurately without leaving partial outer rings. Use a thumbs-up check: 'Show me thumbs up if your table agrees with the Ne and Ar mats.' Record groups needing support with counting the outer level, especially distinguishing the full inner levels from the outer one.

	mark which atoms need the fewest electrons moved.	and Mombasa is a Cl atom looking to gain one electron.' Make sure learners record without yet explaining.		
Explain Phase  VIDEO: Valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Counting valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to their predictions and use Claim-Evidence-Reasoning (CER) to write a short explanation on a sticky note for one question: 'Why does sodium lose one electron rather than gain seven, and why does chlorine gain one rather than lose seven?' Claim: statement about what each atom does. Evidence: data from their table (outer electrons, electrons to lose or gain). Reasoning: link to the stable noble gas arrangement (full outer level, octet; He duplet). They then explain, in whole-class discussion, why Ne and Ar do not react (already full outer level) while Na and K react vigorously (one electron to lose to reach a stable arrangement). They also tackle O: gains 2 to reach Ne-like arrangement. They revise their original prediction in a different colour pen and note what changed.	Say: 'Now you have data. Use it. A good explanation has a claim, evidence from your table, and reasoning about stability.' Show a CER frame on the board. Give 4 minutes of individual writing. Ask: 'Why is it easier for sodium to lose one electron than gain seven?' WAIT TIME: 10-15 seconds. Cold-call 4 learners in turn (Wanjiku, Otieno, Amina, Kipchoge as examples): one on Na, one on Mg, one on O, one on Cl. Probe with: 'What does the atom end up looking like?' and 'How does this connect to neon and argon not reacting?' Introduce vocabulary on the board: stable, full outer level, octet (8 outer electrons), duplet (He, 2). Correct misconceptions: 'Atoms do not literally want things; they become more stable with a full outer level.' Say: 'Atoms in the same group, like Li, Na and K, all have one outer electron to lose, so they behave alike even though their masses differ.' Do NOT draw dot-and-cross diagrams or write ion charges in detail; say: 'Tomorrow we will name these charged particles and write their charges.'	Claim-Evidence-Reasoning (Constructing Explanations, SEP 6) combined with whole-class Talk Moves (revoicing, 'who can add on?'): writing a structured explanation forces learners to connect data to the concept of noble gas stability, while discussion lets them refine and challenge each other's reasoning and revisit their initial predictions.	Indicators of understanding: (1) the claim states that Na loses 1 and Cl gains 1 electron; (2) evidence cites the table numbers (e.g. 'Na needs 1 move to lose but 7 to gain'); (3) reasoning uses 'full outer level', 'octet' or 'like Ne/Ar'; (4) learners can state He is stable with 2 electrons (duplet). Collect sticky notes and sort into three piles: complete CER, partial (missing reasoning), and misconception. Exit check: ask all learners to hold up 1, 2, 3 or 4 fingers for the number of outer electrons Mg needs to lose to reach a stable arrangement (answer: 2 fingers).
Driving Question Board (DQB) Creation  VIDEO: Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum)	Learners return to the class Driving Question Board (DQB) started in Lesson 1. In groups, they review existing questions and place a green dot on any that today's lesson helps answer (e.g. 'Why do Ne and Ar not react?', 'Why do Li, Na and K react alike?'). They write one or two new questions on sticky notes arising	Say: 'Walk to the DQB with your group. Find a question we have partly answered today.' Give 2 minutes. Ask: 'Which question can we now answer with evidence? Which new question has come up?' WAIT TIME: 10-15 seconds before taking responses. Cold-call 3 groups to read one question aloud and say how the stability	Driving Question Board questioning routine: returning to a public record of questions lets learners see their learning progress, notice what remains unanswered and see the path to the driving question, building ownership of the inquiry.	Look for: (1) at least one correctly matched DQB question answered with 'full outer level' reasoning; (2) new questions that move beyond 'why' to predicting charges or formulae (e.g. 'what ion does Mg make?'); (3) every group contributes at least one question; (4) learners listen to and build on peers' ideas without interrupting.

 Search ARES for similar videos  EXERCISE: Counting valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	from today, such as 'What exactly is the charge when sodium loses an electron?', 'Do all atoms reach a full outer level by losing or gaining?' or 'How does chlorine make Lake Victoria water safe?'. They place new notes on the DQB under 'Stability' or 'Ions' and talk briefly about how they relate to the driving question.	idea helps answer it. Sort new questions into themes: 'What are the charged particles formed?' (Lesson 6), 'How many electrons does each element lose or gain?' (Lesson 7), 'How do atoms combine?' (Lessons 8 and later). Say: 'Great questions. Tomorrow we tackle the first of these.' Praise respectful listening to support the Peace value.		Record the number of groups producing questions that link to ion formation as a signal of readiness for Lesson 6.
Model Building Phase  VIDEO: Valence Electrons and the Periodic Table - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Counting valence electrons Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners gather around the Class Model poster, the 'Electron Arrangement Model'. Using the model from Lessons 3 and 4 (group = outer electrons; period = occupied energy levels; family reactivity), each group suggests one addition for stability. They add: (1) a new box 'Stable arrangement: full outer level (octet 8; duplet 2 for He) as in He, Ne, Ar'; (2) an arrow showing 'few outer electrons (Na 1, Mg 2): tend to lose electrons'; (3) an arrow showing 'nearly full outer level (O 6, Cl 7): tend to gain electrons'; and (4) a note linking to the phenomenon: 'Ne and Ar already stable, so no reaction'. They also note a gap: 'We still need to name the particles formed and their charges' as a Lesson 6 task. Learners copy the updated model into their notebooks and label with a Kenyan example, such as chlorine in Nairobi and Mombasa water treatment.	Say: 'A model is a tool to explain and predict. What do we need to add today so our model can explain why Ne is unreactive?' Give 2 minutes for groups to draft their additions on mini whiteboards. WAIT TIME: 10-15 seconds before calling groups. Cold-call 3 groups to propose one addition each; ask the class 'Do you agree? What evidence supports it?' Invite two learners to write the agreed additions on the poster. Ask: 'Does our model yet explain why Li, Na and K react alike?' (yes: same outer electron, same route to stability) and 'Can it predict the charge of the particle formed?' (not yet). Close by pointing to the unanswered gap: 'Tomorrow in Lesson 6 we will predict the ions and draw dot-and-cross diagrams.' Remind them to bring their Element Cards.	Cumulative model revision (Developing and Using Models, SEP 2): adding the stability idea to the existing Class Model shows how new evidence refines earlier ideas, and identifying gaps keeps the inquiry moving toward predicting ions and compounds.	Indicators: (1) the group proposal correctly states that atoms lose or gain electrons to reach a full outer level like a noble gas; (2) learners link the Class Model to both Ne/Ar unreactivity and the Na/K reactivity; (3) learners identify at least one limitation (cannot yet predict charges). Use a 3-question exit ticket: 'Why is neon unreactive? Will Mg lose or gain electrons and how many? Which noble gas arrangement does Cl reach?' Expected: full outer level; lose 2; Ar-like (2,8,8). Review tickets before Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners use evidence from the counter table rather than saying atoms 'want' electrons? Which wording did they use? (2) How many groups correctly identified the easier route (lose 1 for Na, gain 1 for Cl, lose 2 for Mg, gain 2 for O)? (3) Which learners confused total electrons with outer electrons, and what support (colour-coded counters for inner and outer levels) might help? (4) Did the CER frame help quieter learners such as those who rarely speak? (5) Were the cold-calls and WAIT TIME of 10-15 seconds spread fairly across the class? (6) Which DQB questions show readiness for ion formation in Lesson 6? Note adjustments to

timing, and prepare the Lesson 6 examples (Na⁺, Mg²⁺, O²⁻, Cl⁻).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Ne (2,8), Ar (2,8,8) and He (1s ²) have full outer levels and do not react. Na (1 outer electron) and Mg (2) can reach a full outer level by losing few electrons; O (6) and Cl (7) can reach it by gaining few electrons. After moving electrons, each atom had the same arrangement as a noble gas.
What did I learn?	A full outer level (octet of 8, or duplet of 2 for He) is a stable arrangement. Atoms with few outer electrons tend to lose them, and atoms with nearly full outer levels tend to gain electrons, so they become stable like noble gases. Atoms in the same group, such as Li, Na and K, behave alike because they have the same outer electron count.
How does this explain the phenomenon?	Ne and Ar do not react because their outer levels are already full, whereas Na and K react vigorously because losing one outer electron gives a stable arrangement, and chlorine used in Nairobi and Mombasa water treatment reacts because it needs just one more electron to reach the Ar arrangement. This helps explain why elements in the same group behave alike and why neighbours such as Na, Mg, Ne and Ar differ.

LESSON 6 (40 minutes (one double lesson of 80 minutes may be used for the Model Building Phase)): Ion Formation: Predicting Cations and Anions and Drawing Dot-and-Cross Diagrams

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the electron arrangements of atoms and the noble gas target from Lesson 5 to predict whether an atom forms a cation or an anion, draw dot-and-cross diagrams of the ions, and write the electron arrangements of the ions in s and p notation. This advances the driving question by showing how outer electron count lets us predict ions.
Knowledge	Learners will be able to: (a) relate the position of an element in the periodic table to its electron arrangement and use it to predict the ion it forms; (b) illustrate ion formation of elements (Na^+ , Mg^{2+} , Al^{3+} , O^{2-} , F^- , Cl^- , K^+ , Ca^{2+}) using dot-and-cross diagrams; (c) write the electron arrangement of an ion in s and p notation and state which noble gas it matches (Na^+ , Mg^{2+} , O^{2-} , F^- match Ne; Cl^- , K^+ , Ca^{2+} match Ar).
Skills	Developing and using models (dot-and-cross diagrams); analysing and interpreting data (comparing atom and ion electron arrangements); using mathematics and computational thinking (counting electrons and charges); constructing explanations; engaging in argument from evidence.
Attitudes	Learners appreciate the role of electron arrangement in the development of the periodic table. Core competencies: Critical thinking and problem solving (creating new ideas while drawing diagrams of ions using dots and/or crosses to illustrate ion formation); Communication and Collaboration (listening keenly and using open questions while discussing with peers). Value: Peace (showing compassion by displaying tolerance while predicting the type of ion formed from the electron arrangement of elements with peers). PCIs: Life skills (analytical thinking) as the learner predicts the type of ion formed from a given electron arrangement of an atom; Socio-Economic and Environmental Issues (social cohesion) as the learner discusses with peers how atoms acquire stability.
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? (Today's focus: Which ion will an atom form, and how does the ion's electron arrangement compare with the noble gases Ne and Ar?)
Purpose in Storyline	Lesson 5 explained that atoms lose or gain electrons to reach the stable full outer level of noble gases. Lesson 6 turns that idea into predictions: learners work out the exact ion each atom forms, its charge and its new electron arrangement. This gives the ion charges that Lesson 7 (valency and oxidation numbers) and Lesson 8 (formulae) will build on, and it explains why Na and Cl in water treatment and salt behave as they do.
Safety Notes	This is a paper-and-pencil and modelling lesson, so there are no chemical hazards. If a teacher demonstration is used to recall the Na or K reaction with water, use only a video or the teacher's earlier notes; do not let learners handle reactive metals. Use coloured counters or beads carefully (do not put them in the mouth) and keep the floor clear of dropped items.

B. LESSON OVERVIEW

Mwalimu Achieng recalls the puzzle from the Kisumu class: sodium reacts vigorously with water while neon does not react at all. Learners first predict what happens to a sodium atom (2,8,1) and a chlorine atom (2,8,7) when each moves towards a stable arrangement. They then work in groups of four (Wanjiku, Otieno, Amina and Kipchoge style roles) with counters, dot-and-cross cards and the Element Cards to build ions for Na, Mg, Al, O, F, Cl, K and Ca. They write each ion's electron arrangement in s and p notation and discover that Na^+ , Mg^{2+} , O^{2-} and F^- have the same arrangement as Ne ($1s^2 2s^2 2p^6$), while Cl^- , K^+ and Ca^{2+} have the same arrangement as Ar ($1s^2 2s^2 2p^6 3s^2 3p^6$). In the Explain Phase learners build a rule: metals with 1 to 3 outer electrons lose them to form cations, and non-metals with 5

to 7 outer electrons gain electrons to form anions. The DQB is updated and the Electron Arrangement Model is revised to include ions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at two Element Cards: sodium (Z11, 2,8,1, 1s2 2s2 2p6 3s1) and chlorine (Z17, 2,8,7, 1s2 2s2 2p6 3s2 3p5). They recall that sodium reacts vigorously with water, that Ne and Ar do not react, and that chlorine is used in Nairobi and Mombasa to treat water. Individually they write a prediction in their notebooks: 'What will a sodium atom do with its outer electron, and what will a chlorine atom do, to become like a noble gas? What charge will each have?' They draw a quick sketch of the electron shells and share with a partner. (3 minutes writing, 2 minutes pair share.)	Display the two cards on the board. Say: 'Yesterday we learned that Ne and Ar are stable because their outer levels are full. Sodium has 1 outer electron and chlorine has 7. Predict: what will each atom do to become stable, and what charge will it carry? Draw it, do not just say it.' Give WAIT TIME of 10-15 seconds of silence before learners write. Circulate and note predictions without correcting. Cold-call 3 learners (e.g. Wanjiku, Otieno, Amina) to share their prediction, and ask 'What made you think that?' Record 2-3 different predictions on the board, including incorrect ones (for example 'Na gains 7 electrons'), and tell learners 'We will test these ideas with evidence in a few minutes.'	Predict-Observe-Explain (POE) with elicitation of initial ideas: writing a prediction and a drawn sketch before instruction makes learners commit to a model, exposes misconceptions about whether atoms 'lose' or 'gain' the smaller number of electrons, and creates a reason to look carefully at the evidence in the Observe Phase.	Look for: (1) learners who link the outer electron count (1 and 7) to the noble gas target from Lesson 5; (2) learners who include a charge in the prediction; (3) the misconception that atoms always gain electrons or that the nucleus changes. Tally on a class list how many predict Na loses 1 electron and Cl gains 1 electron. Accept all predictions at this stage and record the tally for comparison after the Observe Phase.
Observe Phase  VIDEO: Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cation Formation (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of four, learners receive an 'Ion Evidence Pack' with coloured counters (dots for one atom, crosses for the other), an electron-shell template and a data table of atoms: Na (2,8,1), Mg (2,8,2), Al (2,8,3), O (2,8,6 i.e. 1s2 2s2 2p4), F (2,7), Cl (2,8,7), K (2,8,8,1) and Ca (2,8,8,2). Step 1: for Na, they place counters on the outer shell, then move the outer electron away and record the number of protons (11) and electrons (10), the resulting charge (1+) and the new arrangement (2,8). Step 2: they repeat for Cl, adding one counter to the outer	Model one complete example (Na to Na+) on the board with the group table, showing protons staying at 11 while electrons change from 11 to 10. Say: 'Look at the nucleus. Does the number of protons change? Count again.' Circulate with a checklist. Ask groups: 'How many protons? How many electrons? What is the difference, and what charge does that give?' Use WAIT TIME of 10-15 seconds after each question before taking answers. Cold-call 2 groups to show their Cl- diagram on the board and ask others to find any error. Prompt groups that are	Modelling with manipulatives and data table analysis: moving counters physically represents electron transfer, and the table of protons, electrons and charge lets learners see that charge = protons - electrons. Comparing the ion arrangement with Ne and Ar is pattern recognition that links the new ions to the stable noble gases they learned about in Lesson 5.	Observable indicators: (1) groups correctly count protons and electrons for at least six of the eight ions; (2) dot-and-cross diagrams use dots for one atom and crosses for the other, with square brackets and charges; (3) learners write 1s2 2s2 2p6 for Na+, Mg2+, O2- and F- and 1s2 2s2 2p6 3s2 3p6 for Cl-, K+ and Ca2+; (4) learners do not change the number of protons. Use a quick checklist for each group; give a thumbs-up card for correct tables and ask groups with errors to recheck the electron count.

	shell to make 2,8,8, with 17 protons and 18 electrons, giving 1-. Step 3: they complete the table for Mg^{2+}, Al^{3+}, O^{2-}, F^-, K^+ and Ca^{2+}. Step 4: they write the s and p notation for each ion (e.g. Na^+ 1s² 2s² 2p⁶; Cl^- 1s² 2s² 2p⁶ 3s² 3p⁶) and circle which noble gas it matches. They also draw dot-and-cross diagrams: a sodium atom with one dot in the outer shell transferring to a chlorine atom with seven crosses, showing the final ions in square brackets with charges, $[\text{Na}]^+$ and $[\text{Cl}]^-$ with 8 outer electrons.	stuck: 'Which noble gas is closest to this atom, and how many electrons do you need to move to reach it?' Remind learners that dots belong to one atom and crosses to the other, that brackets and charges go on the final ions, and that only electrons move; protons stay in the nucleus.		
Explain Phase  VIDEO: Noble gas configuration <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cation Formation Practice Source: CK-12  Search ARES for similar readings	Each group studies its completed table and writes a rule in the form: 'Atoms with ____ outer electrons ____ electrons to form a ____ (cation/anion) with charge ____ because ____.' They then use the rule to predict, without counters, the ions formed by Li (2,1), Be (2,2), N (2,5) and S (2,8,6), and check against the group number. Learners explain in a 'claim, evidence, reasoning' format why Na^+ and Ne have the same electron arrangement but are different particles (Na^+ has 11 protons, Ne has 10). They also explain why Mg^{2+} needs two electrons to be lost but F^- needs only one to be gained, and link back to the puzzle: sodium reacts vigorously with water because its single outer electron is easily lost to reach a stable arrangement, while neon does not react because it already has a full outer level. A final class discussion links chloride ions in Nairobi and Mombasa water and in common salt (NaCl) to the idea that chlorine gains one electron.	Ask groups to read their rule aloud and ask: 'What evidence in your table supports that rule?' Use WAIT TIME of 10-15 seconds after the question. Cold-call 4 learners from different groups (for example Kipchoge, Amina, Wanjiku, Otieno) to share and ask others to agree, add on or challenge: 'Do you agree with Kipchoge? What would you add?' Ask the challenge question: 'Na^+ and Ne have the same arrangement. Are they the same thing? Why not?' and guide with: 'Think about protons.' Draw out that metals (1 to 3 outer electrons) lose electrons to form cations, non-metals (5 to 7 outer electrons) gain electrons to form anions, and the charge equals the number of electrons lost or gained. Write the class rule on the board and link to group numbers: group 1 gives 1+, group 2 gives 2+, group 13 gives 3+, group 15 gives 3-, group 16 gives 2-, group 17 gives 1-. Avoid introducing bond types; say 'In sub-strand 1.4 we will study how ions and shared electrons hold atoms together.' Do not teach oxidation numbers or valency in	Claim-Evidence-Reasoning (CER) with a rule-building discussion: learners build a general claim from the table (evidence) and justify it with the stability idea (reasoning), which turns a set of examples into a transferable predictive rule. The challenge question about Na^+ and Ne targets the misconception that the ion becomes the noble gas atom.	Look for: (1) correct predictions of Li^+, Be^{2+}, N^{3-} and S^{2-} with reasons stated in terms of reaching a noble gas arrangement; (2) use of the words cation and anion correctly; (3) a clear statement that the number of protons does not change; (4) learners use the group number to predict the charge. Use a 3-question exit slip: (a) Write the ion formed by Ca and its s and p notation. (b) Is Cl^- a cation or an anion? (c) Which noble gas has the same arrangement as O^{2-}? Success means at least 2 of 3 correct; those who miss go for a short reteach in pairs at the start of the next lesson.

		detail; say 'Next lesson we will give these charges the names valency and oxidation number.'		
Driving Question Board (DQB) Creation  VIDEO: Electron configurations for the third and fourth periods <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Anion Formation (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board started in Lesson 1. Each group writes on sticky notes (a) one question that has now been answered, for example 'Why does sodium react but neon does not?', and moves it to the 'Answered' section with a short evidence statement (for example 'Na loses 1 electron to become Na⁺ 2,8 like Ne'); and (b) one new question, for example 'How do we know the charge to write when making a formula like Al₂O₃?', 'Why do iron and copper form more than one ion?' or 'What holds Na⁺ and Cl⁻ together in salt?'. They place the new notes on the DQB under 'Ions', 'Formulae' or 'Bonding'.	Say: 'Take two minutes with your group. Which of our questions can we now answer, and what new question has today's work raised?' Give WAIT TIME of 10-15 seconds before groups start writing. Cold-call 3 groups to read one note aloud and ask: 'Where on the board does this belong, and which lesson do you think will help us answer it?' Sort the new questions with the class into those answered by Lesson 7 (valency and oxidation numbers, variable oxidation numbers of iron and copper), Lesson 8 (formulae such as NaCl, CaCl₂, Al₂O₃), and sub-strand 1.4 (bonding). Praise good questions and reword vague ones, for example 'Why is the table like that?' becomes 'How does the number of outer electrons decide the ion formed?'	Driving Question Board as a public record of growing understanding: moving questions to 'Answered' with evidence, and posting new questions, makes learners track their progress, gives ownership of what to investigate next, and links today's ions to the upcoming lessons on valency and formulae.	Observable indicators: (1) at least one answered note per group includes evidence from the ion table (such as 'Cl⁻ is 2,8,8 like Ar'); (2) new questions refer to charges, formulae or bonding rather than recall; (3) learners place questions in the correct category. Record which groups ask about transition elements (Fe, Cu) or bonding so you can use those questions as hooks for Lessons 7 and 10.
Model Building Phase  VIDEO: Naming ions and ionic compounds <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Anion Formation Practice Source: CK-12  Search ARES for similar readings	Learners revise the class poster 'Electron Arrangement Model'. In their groups they add a new section called 'Ions' with: (1) a labelled diagram of Na to Na⁺ and Cl to Cl⁻ with electron arrangements 2,8,1 to 2,8 and 2,8,7 to 2,8,8 (s and p notation shown alongside); (2) a small table linking group number to ion charge (group 1: 1+, group 2: 2+, group 13: 3+, group 15: 3-, group 16: 2-, group 17: 1-); (3) a statement of the revised model: 'Atoms lose or gain electrons to get the same arrangement as the nearest noble gas, forming ions with a charge; the number of protons does not change.' Each group adds one example from Kenya, such as K⁺	Say: 'Our model must now explain why sodium reacts and neon does not, and why elements in the same column form ions with the same charge. Add what today's evidence shows.' Give WAIT TIME of 10-15 seconds, then allow 8 minutes of group work. Walk around and ask: 'What did you change from yesterday's model and why?' Cold-call 2 groups (for example the groups with Kipchoge and Amina) to present their revised section in 1 minute each, and ask the audience: 'What evidence supports that part of the model? What is still missing?' Highlight gaps to carry forward: the model does not yet show how charges combine to give formulae (Lessons 7 and 8) or	Cumulative model revision with peer critique: adding ions to the Electron Arrangement Model connects today's evidence to earlier ideas (outer electrons, groups, stability), and the audience critique asks learners to defend model components with evidence, which strengthens explanation of the anchoring phenomenon.	Look for: (1) the revised model shows correct electron arrangements for atom and ion with the proton number unchanged; (2) the group-number to charge table is accurate; (3) the model explains why Na and K (group 1) behave alike by forming 1+ ions; (4) a Kenyan example is linked correctly to an ion. Use a 4-point checklist per group. Collect the notebook summaries and use them to plan the start of Lesson 7.

	from potassium fertilizers (KNO ₃) used in Kericho and the Rift Valley, Cl ⁻ from chlorine in water treatment at Lake Victoria, or Ca ²⁺ in lime used on farms. They then write in their notebooks a short summary: 'What I observed, what I learned, how I can explain it.'	how reactions are written (Lesson 9). Remind learners that peace and tolerance in discussions means listening to every group's reasoning, including those who disagree.		
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D. TEACHER REFLECTION

After the lesson, reflect: Which learners could predict ion charges from group numbers without counters, and which still needed to count electrons? Did any learners think that Na⁺ becomes a neon atom, or that atoms change their number of protons? Which dot-and-cross conventions (dots for one atom, crosses for the other, square brackets and charges) caused most errors? How well did the Kenyan examples (potassium fertilizers in Kericho, chlorine in Nairobi and Mombasa) help learners see why ions matter? Were the 10-15 second WAIT TIME pauses respected, and did quieter learners contribute when cold-called? What will you adjust in Lesson 7 so that the link from ion charge to valency and oxidation number is smooth, and how will you use the DQB questions about iron and copper?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Atoms of Na, Mg, Al, K and Ca lose electrons and atoms of O, F and Cl gain electrons. The ions formed have the same electron arrangement as a noble gas: Na ⁺ , Mg ²⁺ , O ²⁻ and F ⁻ match neon (1s ² 2s ² 2p ⁶), while Cl ⁻ , K ⁺ and Ca ²⁺ match argon (1s ² 2s ² 2p ⁶ 3s ² 3p ⁶). The number of protons stays the same.
What did I learn?	Metals with 1 to 3 outer electrons lose them to form cations; non-metals with 5 to 7 outer electrons gain electrons to form anions. The charge equals the number of electrons lost or gained, and the group number helps predict it. Dot-and-cross diagrams show the transfer of electrons, with the ions in brackets with their charges.
How does this explain the phenomenon?	Sodium reacts vigorously because it easily loses its single outer electron to reach the stable neon arrangement, while neon does not react because its outer level is already full. Chlorine gains one electron to reach the argon arrangement, forming Cl ⁻ . Elements in the same group form ions of the same charge because they have the same number of outer electrons, which helps us predict ions of elements we have never worked with.

LESSON 7 (40 minutes): Valency and Oxidation Numbers: What Do Electron Arrangements Tell Us About Charge?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners infer the valency and the signed oxidation number of an element from its electron arrangement, and to discuss why transition elements such as iron and copper show more than one oxidation number.
Knowledge	By the end of the lesson, the learner should be able to: (a) relate the position of an element in the periodic table to its electron arrangement by using outer electrons to state valency; (b) illustrate ion formation by stating oxidation numbers (Na +1, Mg +2, Al +3, Cl -1, O -2) from electron arrangement; (c) discuss elements with variable oxidation numbers (Fe 2+ and 3+; Cu 1+ and 2+).
Skills	Inferring (valency and oxidation number from s and p notation); analysing and interpreting data (tables of outer electrons and ion charges); developing and using models (revising the Class Model 'Electron Arrangement Model'); constructing explanations; communicating with peers.
Attitudes	Appreciate the role of electron arrangement in the development of the periodic table. Show tolerance and compassion (Peace) while predicting ions with peers. Practise analytical thinking (life skills) when predicting oxidation numbers. Core competencies: Communication and Collaboration (listening keenly and using open questions); Critical thinking and problem solving.
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? (Today: what do the outer electrons tell us about valency and oxidation number, and why do iron and copper not follow a single rule?)
Purpose in Storyline	Lesson 6 taught learners to predict cations and anions from electron arrangement and to draw dot-and-cross diagrams (Na ⁺ , Mg ²⁺ , Al ³⁺ , O ²⁻ , F ⁻ , Cl ⁻ , K ⁺ , Ca ²⁺). Lesson 7 uses these ion charges to separate two ideas: valency (an unsigned number) and oxidation number (a signed number). It also shows the limit of the outer-electron rule with the transition elements iron and copper. This prepares Lesson 8, where learners use valencies, oxidation numbers and radicals to derive formulae.
Safety Notes	This lesson is paper-based and uses data cards and a short observation task. If Mwalimu Achieng shows samples of rust (iron oxide) or copper wire or copper sulfate crystals, learners handle them only with clean dry hands, do not taste or put them near the face, and wash hands afterwards. Keep any copper compounds away from the mouth and eyes. No heating or open flames in this lesson.

B. LESSON OVERVIEW

Mwalimu Achieng opens with the anchoring phenomenon: why do Li, Na and K (masses 7, 23 and 39) behave alike, why are Ne and Ar unreactive, and why does chlorine appear in Nairobi and Mombasa water treatment as Cl⁻? Learners already know from Lesson 6 that atoms form ions with fixed charges. Today they predict, then use Element Cards and a data table to infer valency (how many electrons are lost or gained) and oxidation number (the same number with a sign: + for loss, - for gain). They test the outer-electron rule on Na +1, Mg +2, Al +3, Cl -1 and O -2. Then they meet a puzzle: a Kisumu jua kali metalworker's iron nails and roofing sheets rust to Fe₂O₃ while iron in FeCl₂ behaves differently, and copper forms both Cu⁺ and Cu²⁺. Learners discover that transition elements can lose different numbers of electrons, so outer-electron count alone does not predict a single oxidation number. They add this to the DQB and revise the Class Model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Oxidation state trends in periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners sit in groups of four (Wanjiku, Otieno, Amina and Kipchoge's group is used as an example). On a slip of paper each learner independently predicts and writes: (1) the charge on a sodium ion, a magnesium ion, an aluminium ion, a chloride ion and an oxide ion, using what they remember from Lesson 6; (2) whether iron might form only one ion or more than one, and why. They write one sentence beginning, 'I think valency and oxidation number are ... because ...'. They then share predictions in their group and agree on one group prediction to record on the board.	Greet learners and say: 'Yesterday we predicted ions from electron arrangement. Today we ask a sharper question: can we state a charge with a sign, and does every element follow one rule?' Display on the board: Na 2,8,1; Mg 2,8,2; Al 2,8,3; Cl 2,8,7; O 2,6. Say: 'Write your own prediction first. No talking for two minutes.' Circulate silently and note common ideas. Ask: 'Is valency the same thing as charge? Write what you think.' WAIT TIME 10-15 seconds after each question before taking answers. Cold-call 3 learners (not volunteers) to read their predictions for Mg and Cl. Cold-call 2 more groups for their prediction on iron. Do not correct answers; record all on the board under the heading 'Our predictions'. Link back: 'Remember our phenomenon: Li, Na and K behave alike. Keep that in mind as we predict.'	Eliciting prior ideas (Predict-Write-Share): writing individually before group talk makes every learner commit to an idea, which exposes misconceptions such as 'valency and oxidation number are exactly the same thing' or 'all metals form one ion'. Comparing predictions with later evidence builds conceptual change and ownership.	Check slips for observable indicators: (a) learners link ion charge to outer electrons (e.g. 'Mg has 2 outer electrons so Mg ²⁺ '); (b) learners attach a sign to the charge; (c) learners give any reasoned idea about iron. Tally on a clipboard how many predicted Cl as 1- and how many wrote 7+. Note learners who assume iron has only one ion for the later discussion.
Observe Phase  VIDEO: Oxidation state trends in periodic table Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar	Part A (15 minutes in total for the Observe Phase, 8 minutes here): Each group receives a Data Table of Element Cards with columns: element, electron arrangement (energy-level form and s and p notation), outer electrons, electrons lost or gained, ion formed, valency, oxidation number. Rows given: Na (2,8,1; 1s ² 2s ² 2p ⁶ 3s ¹), Mg (2,8,2), Al (2,8,3), O (2,6), Cl (2,8,7). Learners complete the blanks and notice patterns: Na loses 1 electron, valency 1, oxidation number +1; Mg loses 2, valency 2, +2; Al loses 3, valency 3, +3; O gains 2, valency 2, -2; Cl gains 1, valency 1, -1. Part B: groups	Distribute the Data Tables and say: 'Your job as chemists is to find the rule that links outer electrons to valency and to oxidation number. Use your Lesson 6 ion charges.' Model one row on the board (Na): 'Sodium has 1 outer electron. It loses it. How many electrons were lost? The valency is 1. Because electrons are negative, losing one makes the ion positive, so the oxidation number is +1.' Then release groups. Circulate and ask open questions: 'What do you notice about the number of outer electrons and the valency?', 'Why does chlorine have a minus sign but sodium has a plus?' Pause	Pattern finding in data tables (Noticing and Wondering): learners compare rows to discover that valency equals the number of electrons lost or gained and that the oxidation number carries a sign. Contrasting main-group data (single numbers) with transition data (multiple numbers) creates a productive puzzle that pushes the rule beyond simple outer-electron counting.	Observable indicators: learners correctly fill at least 4 of 5 rows; they state valency without a sign and oxidation number with a sign; they say aloud that metals give positive and non-metals give negative numbers; they highlight iron and copper rows as exceptions. Use a quick tick sheet. If more than a third of learners write Cl as +7, pause and re-model with the Cl arrangement 2,8,7 needing 1 electron to reach the Ar arrangement.

readings	receive a second card set for transition elements: Iron: Fe²⁺ in FeCl₂ and Fe³⁺ in Fe₂O₃ (rust on roofing sheets and nails in Kisumu); Copper: Cu⁺ and Cu²⁺ (CuO). Learners record the oxidation numbers (+2 and +3 for iron; +1 and +2 for copper) and compare these with the single numbers obtained for Na, Mg, Al, Cl and O. They highlight in a different colour where the table breaks the pattern.	WAIT TIME 10-15 seconds after each question. For Part B, say: 'Look at iron. FeCl₂ and Fe₂O₃ are both real compounds. Which numbers can you write in the oxidation number column for iron? Copper?' Cold-call 4 groups to report one row each (Na, Mg, Al, then Cl or O). Then cold-call 2 groups to report what they noticed about iron and copper. Probe: 'Did the outer-electron rule give us +2 and +3 for iron? What do you think is different about transition elements?' Hand out a short statement: 'Transition elements can lose electrons from more than one energy level, so they can form more than one ion.' Keep the explanation brief; do not teach bonding.		
Explain Phase  VIDEO: The periodic table - transition metals <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Changes in Oxidation Number in Redox Reactions (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners work in their groups to construct a written explanation (3-4 sentences) answering: 'How does electron arrangement let us infer valency and oxidation number, and why can iron and copper have more than one oxidation number?' They must use these words: outer electrons, lose, gain, valency, oxidation number, noble gas arrangement, transition element. They then link the explanation to the phenomenon: they write one sentence explaining why Li, Na and K all have oxidation number +1 despite masses 7, 23 and 39, and why Ne and Ar have no oxidation number to predict in this way (full outer level, 8 outer electrons, no tendency to lose or gain). Learners also complete a short application task: state valency and oxidation number for K, Ca, F and N from their arrangements (K 2,8,8,1: valency 1, +1; Ca 2,8,8,2: valency	Say: 'Now explain it in your own words. Use evidence from your table, not just memory.' Provide sentence starters on the board: 'Because the atom has ___ outer electrons, it ___ electrons, so its valency is ___ and its oxidation number is ___.' Circulate and prompt: 'How do you know that is a loss and not a gain?', 'What would make iron different from sodium?' WAIT TIME 10-15 seconds after each question. Cold-call 3 groups to read their explanations aloud while the rest listen with the task 'Add one idea you did not include'. Ask a follow-up to a different learner each time: 'Do you agree or disagree with that group? Why?' Highlight Amina's or Kipchoge's group (as examples if they offer strong reasoning) and ask others to build on it. Address misconceptions explicitly: 'Valency is never negative. Oxidation number has a sign.' Clarify that	Claim-Evidence-Reasoning with peer critique: learners state a claim (valency and oxidation number follow from electron arrangement), cite evidence from the data table (rows for Na, Mg, Al, Cl, O and for Fe, Cu), and give reasoning using the noble gas arrangement. Listening to others and adding ideas builds the Communication and Collaboration competency and shows the limits of a model.	Look for explanations that (a) correctly state valency as a number of electrons lost or gained; (b) assign the right sign; (c) refer to the stable noble gas arrangement; (d) acknowledge that iron and copper have more than one oxidation number because they are transition elements. Use a 3-point check: 3 = all four elements present; 2 = three; 1 = two or fewer. Collect two exit examples from groups for the 4 K, Ca, F, N answers to verify at least 80% accuracy.

	2, +2; F 2,7: valency 1, -1; N 2,5: valency 3, -3). Finally, pairs use Cl- in water treatment at Nairobi and Mombasa: 'What is the oxidation number of chlorine in chloride, and how did you infer it?'	iron has two common oxidation numbers, +2 and +3, and copper +1 and +2, and that for transition elements the outer-electron count alone does not decide which one forms. Conclude: 'The rule works beautifully for main groups; transition elements remind us that atoms are more complex. This is why FeCl ₂ and Fe ₂ O ₃ contain different iron ions.'		
Driving Question Board (DQB) Creation  VIDEO: Valence Electrons and the Periodic Table - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Changes in Oxidation Number in Redox Reactions Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners review the class Driving Question Board started in Lesson 1. In groups they write new questions on sticky notes prompted by today's evidence, such as: 'Why do transition elements lose different numbers of electrons?', 'How do we know which iron ion is in rust?', 'How do we use oxidation numbers to write formulae like Fe ₂ O ₃ and FeCl ₂ ?', 'What other elements have variable oxidation numbers?' They also move previously posted questions that have now been answered (for example 'Why does sodium form Na+?') to an 'Answered' section, with a short note of the evidence. Each group places at least one new question and one answered question on the DQB.	Say: 'Look at the DQB. Which of our questions did today's evidence help to answer? What new questions has it raised?' Give 3 minutes of silent writing, then 2 minutes of group sharing. Cold-call 4 groups to read one new question and say where it belongs on the board (connected to which part of the phenomenon). Ask: 'Which question will Lesson 8 help us answer?' and point to formula questions. WAIT TIME 10-15 seconds before calling on anyone. Highlight any question about formulae and say: 'Next lesson, we will use these valencies and oxidation numbers, together with radicals such as nitrate in KNO ₃ fertilizer used in Kericho, to derive formulae.' Also remind learners of the link to the driving question: how electron arrangement helps predict ions and compounds.	Driving Question Board tracking: documenting answered and new questions makes learning progress visible and keeps the phenomenon at the centre. Learners see that evidence (the table, the iron and copper puzzle) produces both answers and new questions, which models authentic scientific inquiry.	Observable indicators: each group posts at least one question that is specific to today's content (valency, oxidation number, or variable oxidation numbers); questions are testable rather than yes/no; learners correctly identify at least one answered question and cite the evidence. Note groups whose questions show confusion between valency and oxidation number for follow-up at the start of Lesson 8.
Model Building Phase  VIDEO: Groups of the periodic table <i>Related topic: not an exact match for this lesson.</i>	Learners revisit the Class Model poster 'Electron Arrangement Model'. In groups they add today's ideas: (1) next to each group (1, 2, 13, 16, 17), a label giving the usual valency and oxidation number (group 1: +1; group 2: +2; group 13 (Al): +3; group 16 (O): -2; group	Say: 'Our Class Model must explain more than it did yesterday. What new idea do we add and where?' Hand out markers and let each group add one element to the poster (assign Na, Mg, Al, Cl, O, Fe, Cu across groups). Circulate and ask: 'Where on the model	Cumulative model revision: adding valency, signed oxidation number and the transition element exception to the Class Model connects each new piece of evidence to the full causal chain (electron arrangement -> outer electrons -> group -> ion ->	Observable indicators: the revised poster shows outer electrons leading to valency and then to the signed oxidation number; the transition element note is present with correct numbers (Fe +2, +3; Cu +1, +2); at least 80% of notebooks contain a correct

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum) Search ARES for similar readings	17 (Cl): -1); (2) an arrow from 'outer electrons' to 'valency' to 'oxidation number (with sign)'; (3) a boxed note: 'Transition elements (e.g. Fe: +2, +3; Cu: +1, +2) can have more than one oxidation number'. Each group also draws a small labelled sketch for one element, e.g. Mg: 2,8,2 -> loses 2 electrons -> Mg²⁺ (2,8) -> valency 2, oxidation number +2, same arrangement as Ne. Learners then write in their notebooks: 'The model now explains ... but cannot yet explain ...'.	does the sign of the oxidation number come from?', 'Which part of the model needs a warning label for iron and copper?' WAIT TIME 10-15 seconds after each question. Cold-call 3 learners to explain an addition made by another group and say whether they agree. Introduce the 'gap' sentence: 'The model now explains ____ but cannot yet explain ____.' Expected gaps: how to combine ions into formulae and how to use brackets for radicals. Close by tying back to the phenomenon: 'Because Li, Na and K each have one outer electron, each forms +1 and reacts alike; because Cl needs one electron, it forms -1 and is used to disinfect water in Nairobi and Mombasa. Next lesson we use these numbers to write formulae.'	oxidation number). Naming what the model cannot yet explain builds scientific honesty and sets up the next lesson.	'explains/cannot yet explain' sentence. Collect one notebook sentence per group as an exit ticket and use it to plan Lesson 8.
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D. TEACHER REFLECTION

After the lesson, consider: Did learners distinguish valency (unsigned, whole number) from oxidation number (signed)? Which groups still wrote Cl as +7 or O as +2, and what did their reasoning reveal? How well did learners cope with the iron and copper puzzle, and did any learner claim that all metals form one ion? Were all learners participating in the cold-call routine, including quieter learners? Did the Kenya contexts (rusting roofing sheets in Kisumu, chlorine in Nairobi and Mombasa water) help learners connect the ideas to the phenomenon? Note which DQB questions to carry into Lesson 8, such as formulae for FeCl₂ and Fe₂O₃, and how you will revisit unresolved misconceptions at the start of that lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that Na, Mg and Al lose 1, 2 and 3 electrons to give oxidation numbers +1, +2 and +3, while Cl gains 1 and O gains 2 to give -1 and -2. We observed that iron forms Fe²⁺ and Fe³⁺ (in FeCl₂ and Fe₂O₃) and copper forms Cu⁺ and Cu²⁺, so these transition elements have more than one oxidation number.
What did I learn?	We learned that valency is the number of electrons an atom loses, gains or shares (a positive whole number), and the oxidation number is the same number with a sign (+ for loss, - for gain). We learned that outer electrons let us infer these numbers for main-group elements, but transition elements such as iron and copper can have variable oxidation numbers.
How does this explain the phenomenon?	Elements in the same group, such as Li, Na and K, have the same outer electron count, so they all have valency 1 and oxidation number +1 and react alike despite masses of 7, 23 and 39. Atoms lose or gain electrons to reach the stable noble gas arrangement, which is why Na forms +1 and Cl forms -1. Iron and copper show that the outer-electron rule has limits, and this

	prepares us to use valencies and oxidation numbers to derive formulae next lesson.
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LESSON 8 (40 minutes (one lesson; extend to a double lesson if learners need more practice)): Radicals and Deriving Formulae: Predicting the Compounds Elements Form

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the valencies and oxidation numbers inferred from electron arrangement in Lessons 6 and 7, together with radicals, to derive formulae of compounds. This shows how electron arrangement lets chemists predict the compounds an element forms, such as NaCl, CaCl ₂ , Al ₂ O ₃ and the fertilizer KNO ₃ .
Knowledge	By the end of the lesson, the learner should be able to: (c) derive the formulae of compounds. Learners will know that a radical is a charged group of atoms that stays together (NO ₃ ⁻ , OH ⁻ , SO ₄ ²⁻ , CO ₃ ²⁻ , NH ₄ ⁺ , PO ₄ ³⁻). They will know that a compound is neutral, so total positive charge equals total negative charge. They will know that the criss-cross method swaps the valencies and writes them as subscripts, with brackets around a radical when a subscript is needed, e.g. Ca(NO ₃) ₂ .
Skills	Using mathematics and computational thinking (NGSS SEP 5): applying the criss-cross method and checking the charge balance. Constructing explanations (SEP 6): explaining why Al ₂ O ₃ has the ratio 2:3. Developing and using models (SEP 2): revising the Class Model to link electron arrangement to ion charge to formula. Communication and Collaboration: listening keenly and using open questions in pairs.
Attitudes	Appreciate the role of electron arrangement in the development of the periodic table (outcome e). Show tolerance and patience with peers while checking each other's formulae (Peace). Life skills (analytical thinking): reasoning step by step from electron arrangement to formula.
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form?
Purpose in Storyline	Lesson 7 gave the valency and signed oxidation number of each element from its electron arrangement. Lesson 8 takes the next step by combining those ions into neutral compounds, so learners can predict formulae such as NaCl, CaCl ₂ , Al ₂ O ₃ and KNO ₃ without having seen them. This prepares for Lesson 9, where the formulae are used in balanced equations, and for Lesson 10, where learners predict SrCl ₂ and SrO for strontium.
Safety Notes	This is a paper-and-card lesson with no chemicals. If a real fertilizer sample (e.g. KNO ₃ or ammonium sulfate) is shown, keep it in a sealed container, do not let learners taste or handle it, and have learners wash their hands afterwards. Remind learners that fertilizers are not food.

B. LESSON OVERVIEW

Mwalimu Achieng opens with the phenomenon. Chlorine compounds treat water in Nairobi and Mombasa, and potassium compounds such as KNO₃ feed crops in Kericho and the Rift Valley. How do chemists know the formula of such compounds? Learners predict the formulae of NaCl, CaCl₂ and Al₂O₃ from the ion charges they know. They then use Ion Cards (Na⁺, Mg²⁺, Ca²⁺, Al³⁺, K⁺, Fe²⁺, Fe³⁺, Cu²⁺, Cl⁻, O²⁻) and Radical Cards (NO₃⁻, OH⁻, SO₄²⁻, CO₃²⁻, NH₄⁺, PO₄³⁻) to build neutral compounds, testing the charge-balance rule and the criss-cross method. Learners then derive the fertilizer formulae KNO₃, K₂SO₄, Ca(NO₃)₂ and (NH₄)₂SO₄ and the formulae with variable oxidation numbers (Fe₂O₃, FeCl₂, CuO). They explain the patterns and add 'electron arrangement -> ion charge -> formula' to the Class Model and the DQB. Each of Wanjiku, Otieno, Amina and Kipchoge from the Kisumu class appears in the worked examples.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Oxidation state trends in periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the Kisumu class phenomenon: Lake Victoria water is treated with chlorine and farms in Kericho use KNO ₃ . Each learner silently writes a prediction on a sticky note or in the notebook: 'Sodium forms Na ⁺ and chlorine forms Cl ⁻ . What is the formula of the compound they make, and why?' Then: 'Aluminium forms Al ³⁺ and oxygen forms O ²⁻ . Predict the formula of aluminium oxide and explain how many of each ion are needed.' They write one sentence of reasoning and draw a quick sketch of the ions. Typical predictions are NaCl and AlO (a common error), which are used as evidence later.	Display the question on the board and say: 'Wanjiku, Otieno, Amina and Kipchoge are told that sodium forms Na ⁺ and chlorine forms Cl ⁻ . What compound do they make?' Give WAIT TIME of 10-15 seconds in silence, then 2 minutes of writing. Say: 'Do not worry about being right; I want your best thinking from Lesson 7.' Cold-call 4 learners to read predictions without commenting on right or wrong, and record them in two columns on the board: 'AlO' and 'Al ₂ O ₃ ' (and any other ideas). Ask: 'What is your reasoning for the number of ions?' Hold up a 1 cm piece of paper labelled '+' and '-' to show the charges.	Eliciting prior knowledge through individual prediction (Predict-Observe-Explain). Writing before talking makes each learner commit to an idea. The class can then compare their predictions with evidence, which exposes the common error of ignoring charge balance.	Check the notes. Indicators: (1) learner correctly names Na ⁺ and Cl ⁻ and gives NaCl; (2) learner states the charge on Al ³⁺ and O ²⁻ correctly but not the ratio, which is the expected gap; (3) learner mentions that charges must 'cancel' or 'balance' without being prompted. Tally correct NaCl and Al ₂ O ₃ predictions. Note learners who write AlO or Al ₃ O ₂ for follow-up in the Observe Phase.
Observe Phase  VIDEO: Oxidation state trends in periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Changes in Oxidation Number in Redox Reactions	In groups of 4 (about 8 groups), learners get a pack of Ion Cards (Na ⁺ , K ⁺ , Mg ²⁺ , Ca ²⁺ , Al ³⁺ , Fe ²⁺ , Fe ³⁺ , Cu ²⁺ , Cl ⁻ , O ²⁻) and Radical Cards (NO ₃ ⁻ , OH ⁻ , SO ₄ ²⁻ , CO ₃ ²⁻ , NH ₄ ⁺ , PO ₄ ³⁻). Station 1: they build NaCl, MgO, CaCl ₂ and Al ₂ O ₃ by placing cards side by side until the total charge is zero, and record the number of each ion, e.g. Ca ²⁺ with 2 Cl ⁻ gives CaCl ₂ . Station 2: they add the radicals and build KNO ₃ , K ₂ SO ₄ , Ca(NO ₃) ₂ and (NH ₄) ₂ SO ₄ , and notice that brackets are needed when more than one radical is used. Station 3: they build Fe ₂ O ₃ , FeCl ₂ and CuO using the oxidation numbers from	Say: 'You are chemists in Kericho deciding what fertilizer to make. Build each compound so that the total charge is zero.' Circulate and ask: 'How many Cl ⁻ do you need to balance one Ca ²⁺ ?' and 'Why did you put brackets around NO ₃ ?' For groups who write Al ₃ O ₂ , ask: 'Count the charge. What is 3 x (+3)?' Give WAIT TIME of 10-15 seconds after every question before supporting. Do not give the criss-cross rule yet. Stop the class after 15 minutes and cold-call 3 groups to report one compound each; write the results in a class table on the board: formula, ions, ratio. Ask: 'What do you notice about the numbers in the formula	Gathering evidence with physical models (card building) and pattern finding (SEP 4: analysing and interpreting data). Learners build neutral compounds and see the ratio of ions as a consequence of ion charge, which is a consequence of electron arrangement. The pattern is noticed before the rule is named.	Observe whether groups: (1) balance the charge before writing the formula; (2) place brackets around NO ₃ in Ca(NO ₃) ₂ and around NH ₄ in (NH ₄) ₂ SO ₄ , but not in KNO ₃ or K ₂ SO ₄ ; (3) get 2:3 for Al ₂ O ₃ and 1:1 for MgO and NaCl; (4) use Fe ³⁺ for Fe ₂ O ₃ and Fe ²⁺ for FeCl ₂ . Use a checklist with 8 group names. Groups with 3 of 4 correct are ready for the Explain Phase; others get a guiding question.

(Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Lesson 7 (Fe³⁺ with O²⁻, Fe²⁺ with Cl⁻, Cu²⁺ with O²⁻). They record each result in a table: ion 1, ion 2, number of each, formula, total charge. They then compare the numbers in each formula with the ion charges and describe any pattern they notice (the number of one ion equals the charge of the other).	and the charges on the ions?' Record the observations without explaining them.		
Explain Phase  VIDEO: Oxidation state trends in periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Changes in Oxidation Number in Redox Reactions Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners review the class table and explain in pairs why the formulae have the ratios they have, using electron arrangement. They then are introduced to the criss-cross method: write the symbols with their valencies, swap the numbers and write them as subscripts, simplify if possible, and use brackets around radicals when the subscript is greater than 1. They practise: Al³⁺ and O²⁻ gives Al₂O₃; Ca²⁺ and Cl⁻ gives CaCl₂; K⁺ and NO₃⁻ gives KNO₃; Ca²⁺ and NO₃⁻ gives Ca(NO₃)₂; NH₄⁺ and SO₄²⁻ gives (NH₄)₂SO₄. They test it on Mg²⁺ and O²⁻, which criss-crosses to Mg₂O₂ and simplifies to MgO, and on Fe³⁺ and O²⁻ (Fe₂O₃). Each learner writes a 3-sentence explanation: 'Al₂O₃ contains 2 Al³⁺ and 3 O²⁻ because...' Then they complete 6 practice formulae independently: K₂SO₄, Ca₃(PO₄)₂, Mg(OH)₂, Na₂CO₃, CuO and FeCl₃. Finally they link back to the phenomenon: why do Li, Na and K, with very different masses, form similar compounds (LiCl, NaCl, KCl)? Because each has 1 outer electron and forms a 1+ ion.	Say: 'Turn to your partner and explain why aluminium oxide is Al₂O₃ and not AlO. Use the word charge.' Give WAIT TIME of 10-15 seconds, then pairs talk for 2 minutes. Cold-call 3 pairs. Then teach the criss-cross method with the board model: 'Write the ion, swap the numbers, drop the signs.' Emphasise the warning: 'Criss-cross is a shortcut for charge balance, not a replacement for it. Always check that the total positive equals the total negative.' Ask: 'Why do we use brackets in Ca(NO₃)₂ but not in KNO₃?' Pose a challenge: 'What is the formula of calcium phosphate? Predict before we use the method.' Cold-call 4 learners to give the answer Ca₃(PO₄)₂ and check on the board: 3 x 2+ = 6+, 2 x 3- = 6-. Link to the phenomenon: 'Why can a chemist predict the formula of a compound from a group 1 element in any town in Kenya?'	Constructing explanations using the claim-evidence-reasoning (CER) structure with a Think-Pair-Share. Claim: the formula of Al₂O₃. Evidence: the charges on the ions. Reasoning: a compound is neutral and ion charge comes from the electron arrangement. This connects the abstract rule to the underlying cause, which is the electron arrangement and the quest for a stable noble gas arrangement.	Collect the 3-sentence explanations and the 6 practice formulae. Success indicators: 5 of 6 formulae correct, with brackets used correctly in Ca₃(PO₄)₂ and Mg(OH)₂; the explanation mentions that positive and negative charges balance and that charge comes from lost or gained electrons. Common errors to track: changing subscripts inside a radical (e.g. NO₆), forgetting to simplify (Mg₂O₂), and omitting brackets. Use a traffic-light self-rating (green: correct without help, amber: correct with a hint, red: needs more practice).
Driving Question Board (DQB) Creation	Learners return to the DQB wall chart started in Lesson 1. Each group writes on sticky notes: (a) one question from this lesson that	Say: 'Look at the DQB. Which of our questions can we now answer with evidence from today?' WAIT TIME of 10-15 seconds. Cold-call	Question tracking with the Driving Question Board. Moving questions from 'open' to 'answered' with evidence makes learning visible	Check whether the sticky notes show: (1) at least one answered question linked to specific evidence (e.g. Al³⁺ and O²⁻ give

 VIDEO: Oxidation state trends in periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	is now answered, e.g. 'How do we know the formula of potassium nitrate in fertilizers?'; (b) one new question, e.g. 'Why do atoms in NaCl stay together?' (the link to Sub-Strand 1.4), 'What is the formula if the element has two oxidation numbers?', or 'How can we write what happens when sodium reacts with water as an equation?'. They place answered questions in the 'Answered' column with the evidence (the formula, the charge balance) and place new questions on the board under the headings 'Formulae', 'Bonding' and 'Equations'.	3 learners to move one sticky note and justify it. Ask: 'What new question has today created?' Cold-call 2 more learners. Point out that questions on writing equations will be taken up in Lesson 9, and questions on how atoms hold together in NaCl in Sub-Strand 1.4 (Chemical Bonding). Acknowledge each contribution with 'Thank you, that is a good scientific question.'	and shows that chemistry develops through questions. New questions create a purpose for Lesson 9.	Al₂O₃); (2) new questions that go beyond recall, such as asking why or how; (3) correct sorting under 'Formulae', 'Bonding' and 'Equations'. Count the number of groups with both an answered and a new question (target: 8 of 8).
Model Building Phase  VIDEO: Oxidation state trends in periodic table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Changes in Oxidation Number in Redox Reactions (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12	Learners revise the Class Model poster 'Electron Arrangement Model'. Each group adds to the existing model (electron arrangement -> group and period -> stability -> ion charge -> valency and oxidation number) a new arrow: 'ion charge -> neutral compound formula (charges balance)'. They add a worked example with the element cards, e.g. Ca (2,8,8,2) -> Ca²⁺ and Cl (2,8,7) -> Cl⁻ -> CaCl₂, and one radical example, K⁺ and NO₃⁻ -> KNO₃ (fertilizer in Kericho). They write a short caption answering: 'How does electron arrangement help us predict the compounds an element forms?' They then complete the individual Summary Table.	Say: 'Our model must now explain compounds, not just ions. Where does your new arrow go, and what is it caused by?' WAIT TIME of 10-15 seconds. Each group proposes one addition; cold-call 4 groups. Ask: 'Does the model explain why NaCl, KCl and LiCl have the same pattern, even though Li, Na and K have very different masses?' Expect: same outer electron count -> same ion charge -> same formula type. Ask: 'What does the model not yet explain?' Guide learners to the gaps: how atoms share electrons (Sub-Strand 1.4), and how to write equations (Lesson 9). Mark these on the model as 'To be explained'.	Cumulative model revision (SEP 2: developing and using models). Adding a link from ion charge to formula extends the model's predictive power, so learners can use it for compounds they have not met. Naming the gaps keeps the model honest.	Look for models that: (1) show the causal chain from electron arrangement to ion charge to formula; (2) include at least one worked example with correct ions and formula; (3) mention that the compound is neutral; (4) identify at least one gap. Use the Summary Table entries as an exit ticket: the 'explained' column should state that charge balance, which comes from electron arrangement, decides the formula.

 Search ARES for similar readings				
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners balance charges first or only follow the criss-cross trick? Look at errors such as Mg_2O_2 or Al_3O_2 for evidence. (2) Which learners changed subscripts inside radicals, and do they need a physical card model again? (3) Did the examples with Kericho fertilizers and Lake Victoria water treatment make the link to the phenomenon clear? (4) Were learners able to explain why Li, Na and K form similar compounds despite different masses? (5) What will you adjust in Lesson 9 so learners use today's formulae correctly in balanced equations (e.g. $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$)? Note which pairs worked well together to support peer teaching.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that compounds are neutral, and that ions combine in ratios that balance their charges: NaCl (1:1), MgO (1:1), CaCl_2 (1:2), Al_2O_3 (2:3), KNO_3 and $\text{Ca}(\text{NO}_3)_2$ with radicals, and Fe_2O_3 , FeCl_2 and CuO for elements with variable oxidation numbers.
What did I learn?	We learned that a radical is a charged group of atoms that stays together (NO_3^- , OH^- , SO_4^{2-} , CO_3^{2-} , NH_4^+ , PO_4^{3-}), and that the criss-cross method, with brackets around radicals when a subscript is needed, helps us derive formulae by swapping valencies and checking that the total positive charge equals the total negative charge.
How does this explain the phenomenon?	We can explain that electron arrangement decides the ion charge (for example, the 1 outer electron of K gives K^+ , and 3 outer electrons of Al give Al^{3+}). The ion charges then decide how many of each ion are needed to make a neutral compound, so we can predict formulae such as NaCl, CaCl_2 , Al_2O_3 and KNO_3 and explain why Li, Na and K form similar compounds despite their different masses.

LESSON 9 (40 minutes (one lesson)): Writing and Balancing Chemical Equations: What Happens When Sodium Meets Water?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners write word and symbol equations with correct formulae and balance them by adjusting coefficients only, using the reactions of group 1 metals with water and other simple reactions as evidence of how outer electron count shapes chemical behaviour.
Knowledge	By the end of the lesson, the learner should be able to write balanced chemical equations for simple chemical reactions, including state symbols (s), (l), (g), (aq): $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$, $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$, $2\text{Li} + 2\text{H}_2\text{O} \rightarrow 2\text{LiOH} + \text{H}_2$, $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$, $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ and $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$. Learners will also recognise that the same equation pattern fits Li, Na and K because each has one outer electron and forms a 1+ ion.
Skills	Developing and using models (writing equations as symbolic models of reactions); analysing and interpreting data (counting atoms on each side); using mathematics and computational thinking (balancing by coefficients); constructing explanations; communicating and collaborating when peers check each other's equations.
Attitudes	Appreciate the role of electron arrangement in the development of the periodic table; show precision, patience and tolerance when checking peers' work; value the contribution of scientists in representing reactions with symbols.
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? (Today: how can one balanced equation pattern describe Li, Na and K, though their masses differ greatly?)
Purpose in Storyline	Lesson 8 gave learners the skill to derive correct formulae such as NaOH, KOH, NaCl, Al ₂ O ₃ and CaCl ₂ . Lesson 4 gave the observation that Li, Na and K react with water with increasing vigour. In Lesson 9 learners combine these to write balanced equations for those reactions, which shows that elements in the same group follow the same equation pattern because they have the same outer electron count. This supplies the final symbolic evidence needed for the Class Model before Lesson 10, where learners predict the position, ion and formula for strontium.
Safety Notes	Do NOT let learners handle sodium, potassium or lithium. Use only a teacher demonstration video or teacher-led demo with a pea-sized piece of sodium, a large trough of water behind a safety screen, learners at least 2 metres back, and goggles on the teacher. Keep tweezers and filter paper for handling; never touch metal with fingers. Store the metal under oil. Have water and a damp cloth ready. If no metal is available, use the video or the reaction description cards. Dilute HCl for the calcium reaction is shown only by the teacher. The hydrogen gas produced is flammable, so keep naked flames away from the demo. Learners wear goggles when near the demo.

B. LESSON OVERVIEW

Mwalimu Achieng revisits the group 1 water reaction from Lesson 4, in which lithium, sodium and potassium react with increasing vigour despite very different masses (7, 23 and 39). Learners first predict what substances form when sodium meets water and whether the number of atoms stays the same. They then observe a short demonstration or video, collect clues (fizzing, a ball that darts, gas, and a solution that turns red litmus blue), and use their Lesson 8 skills to write correct formulae NaOH and H₂. They discover that the unbalanced equation $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$ does not conserve atoms, and balance it using coefficients only: $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$. In groups they then copy the pattern for K and Li and balance other reactions ($2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$, $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$, $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$) with state symbols. The class adds new questions to the Driving Question Board (DQB) and revises the Electron Arrangement Model (Class Model) to show that the same-group metals follow one equation pattern because each has one outer electron, loses it and forms a 1+ ion.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Visually understanding balancing chemical equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Balancing Chemical Equations (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four (for example Wanjiku, Otieno, Amina and Kipchoge's group). The teacher recalls the Lesson 4 observations: lithium fizzes steadily, sodium melts into a ball and darts, and potassium burns with a lilac flame. Each learner silently writes on a sticky note: (1) what two new substances they think form when sodium reacts with water, (2) whether the total number of atoms after the reaction will be more, less or the same as before, and (3) whether lithium and potassium will follow the same pattern as sodium, and why. Groups then draw a quick sketch of the reaction in a beaker, labelling Na, H ₂ O and any products they predict. All predictions go on the board unchanged and are revisited later.	Say: 'In Lesson 4 we saw sodium dart across water. Today we ask what actually forms, and how chemists write it down.' Show the three metal names Li, Na, K with masses 7, 23, 39. Ask: 'These masses are very different. Will their equations look alike or different? Write your prediction and one reason.' Give WAIT TIME of 10-15 seconds of silent thinking before writing. Circulate and cold-call 4 learners from different groups: 'Wanjiku, what product did your group predict? Otieno, why do you think atoms stay the same or change?' Do not correct predictions; say 'Thank you, we will test that.' Record 3 to 4 distinct predictions on the board under 'Our predictions'. Ask groups to hold on to their sticky notes.	Prediction with a reasoning claim (Predict-Observe-Explain): Writing a prediction and a reason before seeing evidence surfaces prior ideas, such as that atoms may disappear in reactions, so the later observation creates a productive conflict that learners must resolve.	Observe whether sticky notes name plausible products (a hydroxide, a gas, or hydrogen) and whether learners give a reason that links Li, Na and K to similarity (for example the same column or the same outer electrons). Record how many learners predict atoms are conserved. Indicator: at least half of learners can state a reason for their prediction about whether Li, Na and K follow the same pattern. Note misconceptions such as 'atoms are used up' for use in the Explain Phase.
Observe Phase  VIDEO: Balancing chemical equation with substitution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Balancing Chemical Equations Practice Source: CK-12  Search ARES for similar readings	Learners observe a teacher demonstration or video of a pea-sized piece of sodium on water in a large trough, then a short clip or description of potassium (lilac flame) and lithium (steady fizzing). Each group completes an observation table with the columns: What we saw, Evidence of a gas, Evidence of a new substance. They note that the metal disappears, the gas is flammable (hydrogen, shown by a lit splint near the gas collected in an upturned test tube, teacher only), and red litmus placed in the final solution turns blue, showing that an alkaline solution forms. The teacher then gives each group a data card: 'Reactants: Na (solid)	Set the safety rules first: 'Everyone stays behind the line. Goggles on. No one touches the metal.' Say: 'Watch and write only what you can observe, not what you think.' Run the demo, pausing after 20 seconds: 'What do you notice? Talk to your partner for 30 seconds.' Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 3 groups to state one observation each and 2 learners to say which observation is evidence of a gas and which of a new substance. Then hand out the data cards and say: 'Use what you learned in Lesson 8 about formulae to check the formulae NaOH and H ₂ . Why is it NaOH and not NaOH ₂ ?' Cold-call 2	Evidence table with claim-evidence pairing and atom-counting: Learners link each observation (gas, alkaline solution, metal disappearing) to a species in the equation, then use an atom inventory to test whether the equation obeys conservation of atoms. This builds the idea that an equation is a model that must match both the evidence and the conservation rule.	Check that observation tables separate observations from interpretations, and that learners write NaOH and H ₂ correctly (H ₂ as a diatomic molecule). Indicator: at least 80 percent of groups produce the correct word equation and the unbalanced symbol equation, and can show on the atom table that hydrogen atoms (and oxygen atoms) are unequal. Listen for learners who say 'the equation is wrong because the numbers do not match'.

	and H ₂ O (liquid). Products: NaOH (aqueous) and H ₂ (gas).' Learners write the word equation: sodium + water -> sodium hydroxide + hydrogen. They then write the unbalanced symbol equation Na + H ₂ O -> NaOH + H ₂ using the Lesson 8 formulae, and count atoms of Na, H and O on each side in a table, finding 2 hydrogen atoms in H ₂ O on the left but 3 on the right (1 in NaOH and 2 in H ₂).	learners to justify using the 1+ charge of Na ⁺ and the 1- charge of OH ⁻ . Build the atom-counting table on the board with learners calling out numbers. Ask: 'Count hydrogen on the left and right. What do you notice?' Wait 10-15 seconds and cold-call 2 learners. Then say: 'Atoms cannot appear or vanish. What does that tell us about our equation?'		
Explain Phase  VIDEO: Visually understanding balancing chemical equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Balancing chemical equations 1 Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners learn the balancing rule: change the coefficients (big numbers in front), never the subscripts. Guided by the teacher, they balance Na + H₂O -> NaOH + H₂ step by step: (1) put 2 in front of NaOH to give an even number of H on the right? Instead, they test options, find that 2 H₂O gives 4 H on the left, then 2 NaOH + 1 H₂ gives 2 + 2 = 4 H on the right, which needs 2 Na on the left. The result is 2Na + 2H₂O -> 2NaOH + H₂, with state symbols: 2Na(s) + 2H₂O(l) -> 2NaOH(aq) + H₂(g). Groups then use the same pattern to write 2K + 2H₂O -> 2KOH + H₂ and 2Li + 2H₂O -> 2LiOH + H₂, and explain why the pattern is the same: each atom has one outer electron, forms a 1+ ion, and combines with OH⁻ in a 1:1 ratio. They then balance three more equations on a worksheet: 2Na + Cl₂ -> 2NaCl (related to chlorine used in water treatment in Nairobi and Mombasa), 4Al + 3O₂ -> 2Al₂O₃ (aluminium surface) and Ca + 2HCl -> CaCl₂ + H₂. For each, they check their coefficients with an atom table and write which formula came from the ion charges (Al³⁺ and O²⁻, so Al₂O₃; Ca²⁺ and Cl⁻, so CaCl₂). Learners compare their answers	Say: 'Our equation fails the atom count. In balancing there is one rule: change the big numbers in front, never the small numbers inside a formula. Why not? Because changing NaOH to NaOH₂ makes a different substance.' Model the first balance aloud with a think-aloud, then say: 'Your turn. Take the potassium reaction and balance it with your partner.' Give WAIT TIME of 10-15 seconds before groups start and again before taking answers. Circulate with the prompt: 'Show me the atom count on each side.' Cold-call 3 learners to explain one step each and 2 learners to explain why Li, Na and K have the same pattern, expecting the answer 'one outer electron, 1+ ion'. For Al and O₂, prompt: 'Use the ions Al³⁺ and O²⁻ to get the formula first, then balance.' Address the misconception that balancing means editing subscripts by writing 'Al₂O₃ not AlO' on the board. Finish by asking: 'Why does the mass of the metal not change the equation pattern?' Cold-call 2 learners and affirm answers linking to outer electrons.	Worked example then guided practice with a 'same pattern' comparison (analogy across group 1): Modelling the first balance, then letting learners transfer it to K and Li, builds the idea that group membership, not mass, predicts the form of the equation. The atom table makes the conservation of atoms visible and supports the argument that equations are models of real reactions.	Collect whiteboards or worksheets. Indicators: learners balance 2K + 2H₂O -> 2KOH + H₂ and 2Li + 2H₂O -> 2LiOH + H₂ without help; they never change subscripts; they include correct state symbols (s), (l), (g), (aq) in at least the sodium equation; and at least two learners per group can explain that Li, Na and K follow the same pattern because of one outer electron. Use a 3-2-1 check: 3 correctly balanced equations is secure, 2 needs support, 1 or fewer needs re-teaching.

	with a partner, then with the board.			
Driving Question Board (DQB) Creation  VIDEO: Balancing chemical equation with substitution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Balancing Chemical Equations Source: PhET  Search ARES for similar readings	Learners return to the DQB started in Lesson 1. In groups they write new questions or answer existing ones on sticky notes, for example: 'Why does potassium react more vigorously than sodium if the equation looks the same?', 'Why does the equation use H₂ and not H?', 'Can we predict the equation for rubidium, which we have never used?', 'How does the electron move when sodium reacts with chlorine in $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$?' Each group moves at least one earlier question to an 'Answered (partly)' area, such as 'What do Li, Na and K have in common?' (same outer electron count, same equation pattern) and 'How do we write the compounds they form?' (formulae from Lesson 8 and now equations). Open questions that point to bonding, for example how atoms share electrons in Cl₂, stay on the board as a link to sub-strand 1.4.	Say: 'Look at the DQB. Which questions can we now answer using today's equations? Which new questions has the balanced equation raised?' Give WAIT TIME of 10-15 seconds for silent reading of the board. Cold-call 4 learners: 'Amina, choose one question we can now answer and tell us the evidence. Kipchoge, give one new question.' Praise questions that go beyond the lesson. Sort questions into 'answered by evidence', 'partly answered' and 'for later (bonding)'. Ask: 'Which question needs ideas about how electrons move or are shared?' Mark it as a link to the Lesson 10 preview and sub-strand 1.4 without teaching bonding now.	Question-tracking with the Driving Question Board: Moving questions between categories and adding new ones makes learners' growing understanding visible and keeps the phenomenon (same column, same behaviour) at the centre of the inquiry.	Count how many groups add at least one new question that connects to electron arrangement (for example about reactivity differences or ion charges in equations). Indicator: each group moves or answers at least one DQB question using evidence from today, such as the balanced sodium equation, and states what evidence supports the answer.
Model Building Phase  VIDEO: The periodic table - classification of elements <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Chemical Equations: Balancing	Learners gather around the Class Model poster 'Electron Arrangement Model'. In groups they add a new panel titled 'Equations as models of reactions'. They draw Na (2,8,1), then Na⁺ (2,8), and show it joining with OH⁻ in a 1:1 ratio to give NaOH, and write $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$ beneath. They add a note that K and Li follow the same pattern, with K more vigorous because it is lower in group 1, and a note that the Ne and Ar group (full outer level) do not appear in such equations because they do not react. Each learner completes a short exit prompt in their notebook: 'Explain	Say: 'Our model must now explain the symbols we write, not just the electrons.' Ask: 'What should we add to the Class Model so that someone who has never seen our lesson can write the equation for potassium?' Give WAIT TIME of 10-15 seconds, then collect ideas from 3 groups and cold-call 3 learners to read their model panels aloud. Press for the links: 'Where in our model do we show why the coefficient is 2 for Na and for NaOH?' (two atoms each lose one electron). Check that the model avoids teaching bond types. Ask: 'Which part of the model still has a gap?' and note that how atoms share electrons in Cl₂ will be	Cumulative model revision with a causal explanation: Adding equations to the Class Model links the invisible cause (one outer electron, loss to form a 1⁺ ion) to the visible symbolic record (balanced equations), so learners can use the model to predict equations for unfamiliar elements in the same group.	Read the exit prompts. Indicators: the learner states that Li, Na and K have one outer electron and form a 1⁺ ion, so their equations share a pattern; the learner states that balancing uses coefficients to keep atoms equal on both sides; the learner writes at least one balanced equation correctly. Look for the model panel showing 2,8,1 changing to 2,8 for Na⁺. Learners who attribute the pattern to mass need follow-up at the start of Lesson 10.

Chemical Reactions (PLIX) Source: CK-12 Search ARES for similar readings	in two sentences why Li, Na and K have equations of the same pattern, and how we know our equation is balanced.' Learners also fill in the summary table for the lesson.	explored in sub-strand 1.4. Collect exit prompts and read two aloud anonymously. Preview Lesson 10: 'Next time you will use the whole model to predict the position, ion and formula of strontium, an element you have never worked with.'		
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners separate observation from interpretation during the sodium demonstration? (2) How many learners tried to change subscripts when balancing, and which questioning moved them to change coefficients? (3) Could learners explain that Li, Na and K follow the same equation pattern because of one outer electron rather than their masses? (4) Were the Al and O₂ and the Ca and HCl equations too demanding, and who needs extra practice with the atom table before Lesson 10? (5) Which DQB questions about reactivity or electron sharing should be highlighted for the Lesson 10 revisit and the link to sub-strand 1.4? (6) Was the safety routine for the metal demo followed, and would a video be better next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that sodium reacts with water by melting into a ball, darting and fizzing, that potassium reacts even more vigorously with a lilac flame and lithium more gently, that a gas (hydrogen) forms and the solution turns red litmus blue. We also saw that the unbalanced equation $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$ has unequal numbers of hydrogen atoms on each side.
What did I learn?	We learned that a chemical equation must have the same number of each kind of atom on both sides, and we balance it by changing coefficients only, never subscripts. We learned to use state symbols (s), (l), (g), (aq) and to write correct formulae first using valencies from Lesson 8.
How does this explain the phenomenon?	Li, Na and K have very different masses but each has one outer electron, so each loses it to form a 1+ ion and follows the same equation pattern, $2\text{M} + 2\text{H}_2\text{O} \rightarrow 2\text{MOH} + \text{H}_2$ (for example $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$). The pattern is decided by the outer electron count, which sets the group, and not by mass. Balanced equations for $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$, $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ and $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$ follow from the ions each element forms.

LESSON 10 (80 minutes (double lesson)): Final Model Revision and Presentation: Using Electron Arrangement to Explain the Periodic Table and Predict Strontium

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To bring together everything learned in Lessons 1 to 9 so that learners can write a Final Explanation of the phenomenon, show how electron arrangement fixes an element's group and period, and use the same reasoning to predict the position, ion and formulae of an unfamiliar element, strontium. The lesson ends with a link forward to Sub-Strand 1.4 Chemical Bonding.
Knowledge	Learners relate the position of an element in the periodic table to its electron arrangement (outer electrons decide the group, occupied energy levels decide the period). Learners illustrate ion formation as atoms losing or gaining electrons to reach a noble gas arrangement. Learners derive formulae of compounds from ion charges. Learners explain why Li, Na and K (A_r 7, 23, 39) react alike while neighbours such as Na, Mg, Ne and Ar differ.
Skills	Constructing explanations from evidence (NGSS SEP 6), developing and using models by comparing the Lesson 1 and final Class Model (SEP 2), engaging in argument from evidence during peer critique (SEP 7), using patterns to make predictions (SEP 4/CCC Patterns), and communicating scientific information orally and in writing (SEP 8).
Attitudes	Learners appreciate the role of electron arrangement in the development of the periodic table and the contributions of Mendeleev (1869) and Moseley (1913). Learners show peace and tolerance by listening to and respectfully challenging peers' predictions, and show analytical thinking as a life skill when predicting the ion formed from an electron arrangement. Core competencies: Communication and Collaboration, Critical thinking and problem solving, Learning to learn, Citizenship.
Key Inquiry Question	How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? (Forward link to Sub-Strand 1.4: How do chemical bonds influence the properties of substances?)
Purpose in Storyline	This is the final lesson of the storyline. In Lesson 1 learners sorted the element cards by mass and met puzzles: Li, Na and K alike, and neighbours very different. Lessons 2 to 9 supplied the pieces (Mendeleev to Moseley and atomic number, electron arrangement, families, stability, ions, valency and oxidation numbers, radicals and formulae, equations). Lesson 10 asks learners to assemble those pieces into one Final Explanation, test it on an element nobody has studied (strontium), complete the DQB, and see what question remains for chemical bonding.
Safety Notes	No chemicals or heating are used. This is a paper, poster and discussion lesson. Learners move around the room to gallery-walk posters, so keep aisles clear of bags. Do not show or demonstrate alkali metal reactions in this lesson; any earlier reaction observations were from video or teacher demonstration only.

B. LESSON OVERVIEW

Mwalimu Achieng returns to the opening puzzle: sodium, potassium and lithium fizz in water while neon and argon do nothing, and chlorine compounds treat water in Nairobi, Mombasa and from Lake Victoria. Learners first predict where an unfamiliar element, strontium (Z38), would sit and what it would form, using only their own reasoning. They then gather evidence from the whole anchor data set: electron arrangements of the first 20 elements, ion data, and strontium's arrangement 2,8,18,8,2 (1s² 2s² 2p⁶ 3s² 3p⁶ 3d¹⁰ 4s² 4p⁶ 5s²). In the Explain phase groups write and present a Final Explanation of the phenomenon, working through Li/Na/K, Na/Mg/Ne/Ar, Cl-, and the Ar 40 versus K 39 mass puzzle. Learners complete the DQB by sorting questions into answered and open (bonding), and finally compare the Lesson 1 Class Model with the final Electron Arrangement Model and write a revised model that includes strontium and previews the octet idea for bonding.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Valence Electrons and the Periodic Table - Overview Source: CK-12  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12  Search ARES for similar readings	Mwalimu Achieng displays a single new element card: Sr, atomic number 38, A_r about 88, with no group or period shown. Individually, learners write for 3 minutes: (1) which column of the periodic table Sr belongs in, (2) what ion it will form, (3) the formula of its compound with chlorine, and (4) one sentence of reasoning, referring to the Electron Arrangement Model. They then compare predictions with a partner (Wanjiku with Otieno, Amina with Kipchoge style pairs) and note agreements and disagreements on a sticky note. They also write one sentence: 'The hidden property of atoms that creates the patterns in the table is _____. This is the same question posed in Lesson 1, answered again with Lesson 1 to 9 knowledge.	Hold up the Sr card and say: 'Nobody in this class has ever worked with strontium. Chemists in Nairobi or Kisumu laboratories could still predict how it behaves. What would you need to know?' WAIT TIME 10-15 seconds before anyone speaks. Say: 'Write first. I want your own thinking before you talk to anyone.' Circulate and do not correct predictions. After pairs talk, cold-call 4 pairs (not volunteers) to share one prediction and its reason; write each prediction on the board under 'Our predictions' without marking right or wrong. Ask: 'Which prediction used a reason about electrons, and which used a reason about mass?' Point to the Lesson 1 Class Model hung beside the DQB and say: 'In Lesson 1 most of us sorted by mass. Keep that in mind.'	Prediction with reasoning (Predict-Observe-Explain). Committing to a written prediction and a reason before seeing data creates a record that learners can revise, making the growth of their thinking visible and giving them a stake in the evidence that follows.	Look for: how many learners give a reason based on electron arrangement or outer electrons versus mass or guesswork; whether any learners already say 'group 2' or 'Sr2+'; whether predicted formulae follow charges (SrCl2 versus SrCl). Record counts on a tally sheet (e.g. 'reason uses outer electrons: ___ of ___'). Use the tally to decide which step of the Observe phase needs the most support. Learners who give no reason are asked: 'What did you notice about Li, Na and K?'
Observe Phase  VIDEO: Counting valence electrons for main group elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar	In groups of four, learners receive a Whole Evidence Pack on one A3 sheet and use their Element Cards. The pack contains: (a) the table of the first 20 elements with Z, A_r, electron arrangement in energy-level form and s and p notation, group and period; (b) the ion data set (Li+, Na+, K+; Be2+, Mg2+, Ca2+; Al3+; N3-, O2-, F-; P3-, S2-, Cl-) with the noble gas matches (Na+, Mg2+, O2-, F- same as Ne; Cl-, K+, Ca2+ same as Ar); (c) the observations of Li, Na and K with water (Li fizzes steadily and floats; Na melts into a ball and darts; K melts and burns with a lilac flame) and Ne and Ar unreactive; (d) formulae NaCl,	Distribute packs and say: 'You are the chemists with all the evidence. Do not look for answers I am hiding. Look for patterns that hold for every element in the pack.' Circulate with three prompt questions: 'How many energy levels does Sr have, counting 2,8,18,8,2?' (five, so period 5); 'How many outer electrons, and what does that say about its group?' (two, group 2, like Mg and Ca); 'Which noble gas arrangement would Sr+2 match?' (krypton-like 2,8,18,8; accept 'the noble gas before it' if Kr has not been named, and note that Kr is Z36, outside the first 20 elements). If a group claims Sr is group 5	Pattern analysis with structured data table (Analyzing and Interpreting Data, NGSS SEP 4; Patterns CCC). Having learners look for patterns that hold across the entire data set, then extend the pattern to Sr, shows them that the rules are general and predictive rather than facts to memorise.	Check each group's Pattern Finder for: Sr outer electrons = 2, group 2, period 5; Sr ion = Sr2+; ion matched to a noble gas arrangement; group read from outer electrons and period from levels (not swapped). Observable indicators: learners point to the 5s2 end of the notation and say 'outer level'; they use Mg and Ca as comparison rows. Common errors to note: counting 3d10 electrons as outer electrons; using A_r 88 to decide the group; writing Sr2- for the ion. Note names of groups needing re-teaching and pair them with a stronger group in the Explain phase.

readings	MgO, CaCl₂, Al₂O₃, KNO₃, K₂SO₄, Ca(NO₃)₂, (NH₄)₂SO₄; (e) the new Sr data: Z38, electron arrangement 2,8,18,8,2 and 1s² 2s² 2p⁶ 3s² 3p⁶ 3d¹⁰ 4s² 4p⁶ 5s². Learners complete a 'Pattern Finder' table with three tasks: 1. Mark the outer electron count and number of occupied energy levels for Li, Na, K, Mg, Ca and Sr, and state the group and period of each. 2. Find the noble gas whose arrangement matches each ion. 3. Use Ca and Mg to forecast what Sr's outer electrons, group and ion will be, and write the matching row for Sr in the table. They also place the Ar (A_r 40) and K (A_r 39) cards in order and say what decides the order.	because it has five levels, ask: 'Does the number of levels give the group or the period?' and WAIT TIME 10-15 seconds. Cold-call 3 groups to state how they found Sr's group and period. Ask all groups: 'Ar has a bigger A_r than K but comes before it. What decides the order?' Cold-call 2 learners; expect 'atomic number, Moseley 1913, not mass as Mendeleev used in 1869'.		
Explain Phase  VIDEO: Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A (20 min): Each group writes a Final Explanation on a poster using a four-box frame. Box 1, ORDER: what decides the order of elements in the periodic table (atomic number, Moseley), how outer electrons decide group and occupied levels decide period. Box 2, SAME COLUMN, DIFFERENT MASS: why Li, Na and K (A_r 7, 23, 39) each have 1 outer electron and react alike with water, with reactivity increasing Li < Na < K; and why neighbours such as Na, Mg, Ne and Ar differ because their outer electron counts differ (1, 2, 8, 8). Box 3, STABILITY AND IONS: why Ne and Ar do not react (full outer level), why Na loses 1 electron to form Na⁺ (2,8 like Ne), why chlorine gains 1 to form Cl⁻ (2,8,8 like Ar), and how this gives NaCl, CaCl₂ and Al₂O₃, including KNO₃ fertilizer used in Kericho and the Rift Valley and chlorine used to treat water in Nairobi, Mombasa and from Lake Victoria.	Say: 'Your Final Explanation must answer the phenomenon question without using the word because alone. Show me the chain: electron arrangement, then outer electrons, then group, then behaviour.' Model the first link only: 'Sodium is 2,8,1. One outer electron means group 1.' Then step back. While circulating, ask: 'If it is mass that mattered, why would Li and K behave alike?' WAIT TIME 10-15 seconds. Ask: 'Why does Mg form Mg²⁺ and not Mg⁺?' Cold-call 3 learners during Part A to explain one box aloud. Make sure the explanation includes how Sr²⁺ is reached, not just stated. During the gallery walk, stand at two posters and listen for explanations that confuse valency with oxidation number; remind: 'valency is a positive whole number; oxidation number has a sign.' Listen for any claim that atoms 'want' electrons and respond: 'What does it mean in	Claim-Evidence-Reasoning with peer critique (Constructing Explanations, NGSS SEP 6; Engaging in Argument from Evidence, SEP 7). Learners build each statement as claim, data from the evidence pack and a reason in terms of electron arrangement, then test it against peers' feedback. Writing the explanation as a cause-and-effect chain (Cause and Effect CCC) shows that electron arrangement is the underlying mechanism.	Use a four-item rubric on each poster (0 to 2 points each, 8 total): (1) states that order is by atomic number and not mass; (2) links outer electrons to group and levels to period; (3) explains Li/Na/K likeness and Na/Mg/Ne/Ar differences by outer electrons; (4) predicts Sr correctly (group 2, period 5, Sr²⁺, SrCl₂, SrO) with a reason using the noble gas arrangement. Success: 6 or more out of 8. Observable indicators: learners use the words 'outer electrons' and 'noble gas arrangement' in speech; critique cards cite evidence from the pack. Misconceptions to log: mass decides group; atoms 'want' electrons; Sr⁺ formed; subscripts changed to balance equations. Collect posters for the record.

	Box 4, PREDICTION: Sr position (group 2, period 5), ion (Sr^{2+} after losing 2 electrons), formulae SrCl_2 and SrO, and the balanced equation $\text{Sr} + 2\text{HCl} \rightarrow \text{SrCl}_2 + \text{H}_2$ written by analogy with $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$ (with state symbols (s), (aq), (g)). Part B (15 min): Gallery walk and peer critique. Each group sends 1 presenter to stay and 3 visitors to rotate every 4 minutes with a 'Two Stars and a Wish' critique card, using the criteria: uses evidence, links outer electrons to group, uses the noble gas target to explain the ion, and predicts Sr correctly. Part C (5 min): Forward link. Learners write one question about how atoms share electrons, for example how chlorine atoms stay together as Cl_2 or why NaCl differs from other substances. The teacher names this as the focus of Sub-Strand 1.4 Chemical Bonding.	terms of the arrangement of the electrons and the full outer level?' After the walk, cold-call 4 groups to share the one improvement they will make to their poster and allow 3 minutes for revisions. For Part C, say: 'In Lesson 5 we saw stability comes from a full outer level. In bonding, atoms that cannot easily lose or gain electrons will share them. That is the next unit; hold your question for the DQB.'		
Driving Question Board (DQB) Creation  VIDEO: Valence Electrons and the Periodic Table - Overview Source: CK-12  Search ARES for similar videos  HTML: Modern Periodic Table: Periods and Groups (Flexbook) Source: CK-12  Search ARES for similar readings	The class gathers at the DQB, which has been growing since Lesson 1. Each group receives 4 to 5 question strips from the board (assigned by the teacher). They sort their questions into three columns: ANSWERED (give the answer in one sentence using electron arrangement and the lesson where it was learned), PARTLY ANSWERED (what is still missing), and NEW/OPEN (questions for chemical bonding, such as how atoms share electrons or how bonds affect properties of substances). Learners add at least one new question strip from the forward-link question written in Part C, and move answered strips to the 'Answered' side with a green dot. They then co-write on a large	Say: 'Let us go back to the questions we asked on Day One. Which ones can we now answer, and which ones will we carry to the next unit?' Hand out strips, give 5 minutes for group sorting, then cold-call 5 groups (one per column or question) to read a strip and give its answer with evidence. For each answer ask: 'Which lesson gave you that evidence?' WAIT TIME 10-15 seconds before accepting a response. Challenge vague answers: 'Say it using outer electrons.' For open questions, ask: 'Why can the periodic table not fully answer this? What do we need to know about how atoms join?' Then pin the forward-link key inquiry question 'How do chemical bonds influence the properties of substances?' on the DQB as the	Driving Question Board review (question-driven sensemaking). Returning to learners' own questions shows how far understanding has moved and turns unanswered questions into a purpose for the next unit, which sustains curiosity and ownership of learning.	Count and record the proportion of strips moved to ANSWERED and check the accuracy of each stated answer. Observable indicators: learners cite lessons (for example 'Lesson 6, Na^+ is like Ne'); answers use outer electrons and not mass; new questions are about sharing or joining of atoms, showing readiness for bonding. Note any strips placed in ANSWERED with a wrong or incomplete answer and correct them immediately at the board. The two-sentence class answer to the driving question is checked against the outcome: order by atomic number; outer electrons decide group; levels decide period; stability explains ions and formulae.

	paper the answer to the main driving question: 'How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form?' in two sentences.	bridge to Sub-Strand 1.4. Praise respectful disagreement to model the value of peace and tolerance, saying: 'Thank you for disagreeing politely with evidence.'		
Model Building Phase  VIDEO: Counting valence electrons for main group elements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predict the charge on monatomic ions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Individually, learners compare the Lesson 1 version of the Class Model (the 'Electron Arrangement Model' poster, then showing sorting mainly by mass and properties, with puzzles marked) with the current version. In pairs they complete a Model Comparison table with two columns, 'Lesson 1 model' and 'Final model', and rows: What decides the order? What explains the same column? What explains neighbours' differences? Why are Ne and Ar unreactive? How do we predict ions? How do we predict formulae? What can we predict about a new element? As a class they then revise the poster: add Sr in group 2, period 5 with 2,8,18,8,2 and the s and p notation ending 5s2; add arrows from outer electrons to group, levels to period, stability to ion charge, ion charge to formula (SrCl₂, SrO, CaCl₂, NaCl, Al₂O₃, KNO₃); add a 'Next: bonding' box showing atoms that share electrons to reach a full outer level (octet rule preview only; no bond types taught). Each learner writes an individual Final Explanation paragraph (6 to 8 sentences) and completes an exit ticket on a new unfamiliar example: 'Predict the group, ion and formula with chlorine for barium (Z56, outer electrons 2) or for gallium, and explain with outer electrons.'	Say: 'Put your Lesson 1 sheet beside today's model. What has changed in how you think? Point to one thing you had wrong or missing.' WAIT TIME 10-15 seconds. Cold-call 4 learners to name one specific change to the model (for example 'mass was not enough; atomic number decides order'). Build the class poster by asking each of 5 pairs to add one revision and justify it with evidence from the pack. Ask: 'Where on this model would we place a bond? Which part is still missing?' Lead learners to see that the model explains why atoms form ions but not yet how atoms that share electrons stay joined. Do not teach bond types; say only 'bonding in Sub-Strand 1.4 will explain this'. Collect exit tickets and read two aloud (anonymously) for the class to check. Close by thanking Mendeleev's and Moseley's work: 'Scientists in 1869 and 1913 built this table without knowing how electrons are arranged; today you used electron arrangement to explain it and predict an element you had never studied.'	Cumulative model revision with before-and-after comparison (Developing and Using Models, NGSS SEP 2). Placing the first model beside the final model makes conceptual change visible, shows which ideas replaced mass-based sorting, and tests the model's predictive power on a new element, which marks it as a useful scientific model.	Check the Model Comparison tables and exit tickets against these indicators: states that order is by atomic number; links outer electrons to group and energy levels to period; explains ion charge through the noble gas arrangement; derives formula from ion charges (SrCl₂, BaCl₂; for gallium Ga³⁺ and GaCl₃ if used). Success: at least 75% of learners predict group and ion correctly with an electron-based reason, and describe at least two specific changes to their model. Learners who miss it receive a follow-up task: compare Mg and Ca then write Sr's ion. Record data for reporting on specific learning outcomes a) to e).

D. TEACHER REFLECTION

After the lesson, check: Did learners use electron arrangement, not mass, as their main reason when predicting Sr? Which part of the chain (electron arrangement, group and period, noble gas target, ion charge, formula) broke down most often? Did the Final Explanation rubric show learners who still believe mass decides group or that atoms 'want' electrons? Which DQB questions were answered with real evidence, and which new questions point toward chemical bonding? Were all learners, including quieter ones, heard through cold-calls and the gallery walk? What adjustments should be made before starting Sub-Strand 1.4 (for example reviewing stability and outer electrons)? Did learners show respect and tolerance when critiquing peers' predictions?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that elements in the same group (Li, Na, K) react alike despite very different masses (7, 23, 39), neighbours such as Na, Mg, Ne and Ar behave differently, and Ne and Ar do not react. We used the first 20 elements' electron arrangements and ion data to see that outer electron count matches group and noble gas arrangements match ion charges. For strontium (Z38, 2,8,18,8,2) we saw two outer electrons and five energy levels.
What did I learn?	We learned that elements are ordered by atomic number (Moseley, 1913), that outer electrons decide the group and occupied energy levels decide the period, and that atoms lose or gain electrons to reach a stable noble gas arrangement. This lets us predict ions, valencies, oxidation numbers and formulae such as NaCl, CaCl ₂ , Al ₂ O ₃ and KNO ₃ . We also learned that sharing of electrons is explained in bonding (Sub-Strand 1.4).
How does this explain the phenomenon?	The hidden property is electron arrangement. Sr has 2 outer electrons and 5 occupied levels, so it is in group 2, period 5. It loses 2 electrons to form Sr ²⁺ with a noble gas arrangement, like Mg ²⁺ and Ca ²⁺ , so its compounds are SrCl ₂ and SrO. The same reasoning explains why Li, Na and K react alike, why Ne and Ar are unreactive, and why chlorine forms Cl ⁻ for water treatment in Nairobi and Mombasa.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.